GYRO-BASED MULTI-IMAGE DECONVOLUTION FOR REMOVING HANDSHAKE BLUR

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Sung Hee Park
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Marc Levoy, Primary Adviser

I certify that I have read this dissertation and that, in my opinion, it is fully adequate in scope and quality as a dissertation for the degree of Doctor of Philosophy.

Mark Horowitz

I certify that I have read this dissertation and that, in my opinion, it is fully adequate in scope and quality as a dissertation for the degree of Doctor of Philosophy.

Stephen Rock

Approved for the Stanford University Committee on Graduate Studies.

Patricia J. Gumport, Vice Provost for Graduate Education

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Abstract

Digital snapshots have become one of the most common ways that people record and share their experiences in everyday life. Most users would like to take high-quality photographs with their mobile phones or tablets. However, even with recent advances in mobile digital cameras, it is still challenging to capture satisfactory images, especially in low-light situations. In low-light situations, long exposures are inevitable to improve the image quality, but, in turn, handshake blur becomes significant. Accordingly, image deblurring techniques that aim to remove this blur have been intensively studied. Nevertheless, most image deblurring methods are brittle and computationally expensive, making them hardly usable in practice.

This dissertation presents a new gyro-based multi-image deblurring system that handles various challenges in real-world situations. Instead of capturing a single long exposure image, a burst of shorter exposure images is captured along with gyroscope data being recorded at the same time. Any camera shake that occurred during the capture is accurately estimated from the gyroscope data and captured images. Finally, all captured images are jointly deblurred to obtain a sharp and noise-free image of the scene.

Because both camera motion and the scene are accurately measured with a better capture method, gyro-based multi-image deblurring is more robust than previous approaches while its computational burden is significantly reduced. To demonstrate the effectiveness and efficiency of the proposed method, image deblurring is applied to full-resolution imagery of natural scenes captured using a mobile camera and an attached gyroscope.

In addition to the implementation, this dissertation discusses a key question that

applies to any multi-image deblurring method: How many images should be captured, and what should their exposure times be? By quantifying the amount of noise amplified by deblurring, an analysis is provided to estimate how well multi-image deblurring can be done with given exposure parameters. Then, an important property of multi-image deblurring is presented: Multi-image deconvolution does not benefit from increasing exposure time beyond a certain threshold. The noise reduction granted by increasing exposure time is mostly offset by the noise amplification incurred by the more extreme deblurring. Based on this analysis, different capture strategies are compared in various photographic situations, demonstrating that multi-image deblurring is a robust approach that works in low-light situations when other methods fail.

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Chapter 1

Introduction

In recent years, digital cameras have become an almost ubiquitous component in mobile devices, making digital photography a part of everyday life. Most users would like to take high-quality photographs with their mobile phones or tablets. However, although the performance of digital cameras in mobile devices has significantly improved because of recent advances in image sensor technology and software processing, it is still challenging to capture satisfactory images, especially in low-light situations.

In such low-light situations, handshake blur has significant effects on image quality when hand-held devices are used. Long exposure times are often inevitable to reduce the amount of image noise, but, in turn, the images may suffer from blur due to handshake occurred during the capture. Because of this trade-off between image noise and camera shake blur, which is illustrated in Figure 1.1, it is not trivial to capture an image that is sharp and clean of noise at the same time.

To overcome this trade-off and generate compelling images, this dissertation presents a gyro-based multi-image deblurring system that combines two techniques: gyroscope-based camera motion estimation and non-blind multi-image deconvolution. This system captures a burst of images while the gyroscope data is recorded at the same time. Then, the camera shake that occurred during the capture is estimated from both the gyroscope data and image data. Finally, based on the estimated camera motion, all captured images are jointly deconvolved to generate a clean and sharp image. Although these two techniques have individually been the subject of previous

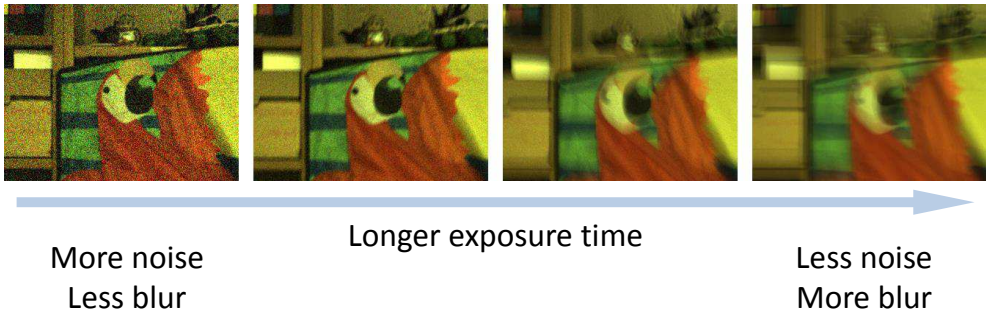


Figure 1.1: A trade-off between image noise and camera shake blur. When a single image is captured in low-light situations, it is challenging to capture an image that is sharp and clean at the same time. When short exposure times are used (shown on the left), the captured image may be degraded by high noise level. On the other hand, an image captured with long exposure times (shown on the right) suffers from camera shake blur.

studies, the combination of them allows us to implement a new image deblurring system that is both effective and efficient enough to be used in practical mobile systems.

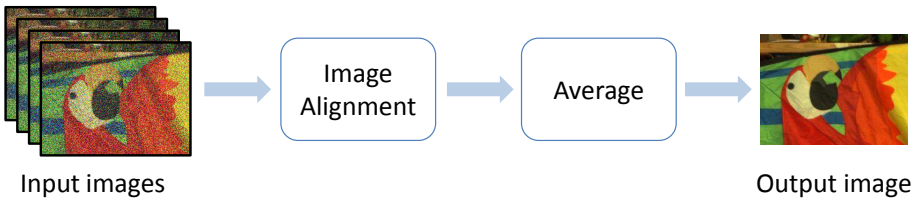
This image deblurring system benefits from two key ideas. First, it adopts a better capture strategy to obtain a good set of images that will be combined after the capture. This multi-image approach provides repeated observations of the scene that preserve more information. Second, a gyroscope is used to directly measure the camera motion. Because it measures both camera motion and the scene as accurately as possible, the estimation becomes more robust than previous image deblurring methods while significantly reducing computational costs.

1.1 Capture Strategies for Removing Handshake Blur

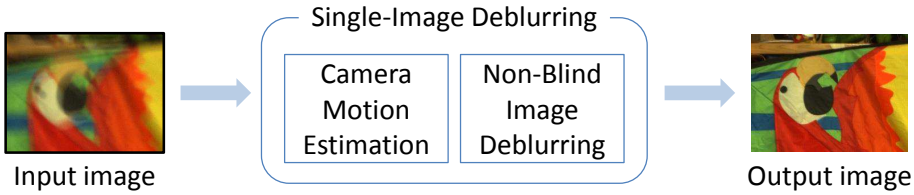
A capture strategy refers to a way of determining a set of capture parameters, such as the number of shots in a burst, exposure time and analog gain used for the burst. This is more general than exposure metering that chooses exposure time and analog gain for a single image capture. Choosing the number of shots becomes an important factor when both noise and handshake blur are involved.

To reduce the amount of blur caused by camera shake, two approaches are commonly used: *align-and-average* and *image deblurring*. Although both approaches aim to solve the same problem, they are based on the two ends of the capture strategy spectrum. On one end is the capture strategy that uses a short exposure time to capture multiple images that are assumed to be blur-free but noisy. The align-and-average method merges these captured images after alignment to reduce noise. On the other end is the strategy of capturing a clean but blurry image using a long exposure time. From this single image, a sharp image is recovered by image deblurring. However, there exists a middle ground between these two ends of capture strategies. A set of images can be captured with intermediate exposure time, hence with some small amount of blur, and jointly deconvolved to recover the latent image. This approach is called *multi-image deblurring*. These approaches are illustrated in Figure 1.2.

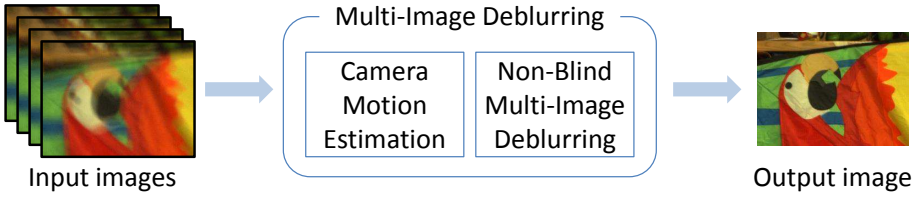
Multi-image deblurring has several advantages over other approaches. Exposure times used in align-and-average must be very short in order to avoid excessive hand-shake blur, and, as noted above, a short exposure time may result in poor image quality that makes alignment difficult and may not even be short enough to completely eliminate hand shake blur. In contrast, multi-image deblurring is applicable to a wider range of exposure times because the blur can be removed computationally, and is thus more effective in low-light situations. In addition, multi-image deblurring is more robust and efficient than single-image deblurring. When multiple images are captured, each image contains different blur caused by different camera motion, and thus preserves frequency components of the scene that are complementary to those captured in the other images. By combining these images, multi-image deblurring significantly improves the quality of output image and reduces ringing artifacts caused by missing frequency components.



(a) Align-and-average



(b) Single-image deblurring



(c) Multi-image deblurring

Figure 1.2: Capture strategies for removing handshake blur. (a) A burst of short exposure shots are captured. The captured images are assumed to be noisy but with little blur. (b) A long exposure is used to capture a blurry and clean image. (c) A set of images are captured with an intermediate exposure time. These images contain some small amount of blur. Note that image-based approaches are assumed for image alignment and camera motion estimation.

1.2 Using a Gyroscope for Removing Handshake Blur

Inertial sensors, in particular accelerometers and gyroscopes, became a must-have component in recent mobile devices. The launch of the Nintendo Wii game consoles and the Apple iPhone brought these sensors into the consumer electronics mass-market [44]. Since then, the uses of micro-electro-mechanical systems (MEMS) inertial sensors expanded rapidly as shown in Figure 1.3. Their manufacturing cost has decreased [16] and these sensors are expected to enable future applications in mobile

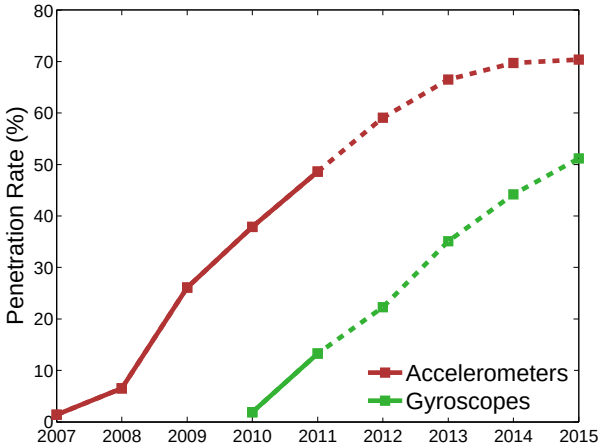


Figure 1.3: Penetration of inertial sensors in mobile phones and tablets. By 2015, Yole Développement [15] estimates that more than half of all mobile phones and tablets are expected to have gyroscopes.

devices such as user interface control, image and video stabilization and augmented reality.

Recent progress in image deblurring was enabled by the use of inertial sensors [30, 55, 7, 51]. A gyroscope measures the rotational component of motion, while an accelerometer records translational motion of the sensor. These measurements eliminate the problem of camera motion estimation, and significantly improve image deblurring. As Figure 1.4 illustrates, blind deblurring requires estimating the camera motion and the scene at the same time. This is an ill-posed problem that consists of two tightly connected subproblems, each of which is challenging. Although it can be solved by an iterative optimization method that alternates between these two subproblems, blind deblurring is considered brittle and computationally heavy because the optimization heavily relies on complicated image prior models to obtain a plausible result. In contrast, if the camera motion is known by the measurements from the sensors, image deblurring becomes non-blind and more tractable because non-blind deblurring is solved only once.



(a) Blind estimation (b) Non-blind estimation

Figure 1.4: Comparison between blind and non-blind deblurring. Blind deblurring is an ill-posed problem that requires solving iteratively two tightly connected subproblems. This is considered brittle and computationally heavy. In contrast, if camera motion is known, the problem becomes non-blind and more tractable because non-blind deblurring is solved only once.

1.3 Comparing the Performance of Different Capture Strategies

A key question about any multi-image deblurring method is: for a given total capture time, how many images should be captured, and what should their exposure times be? Answering this question requires a method that estimates how well deblurring performs in a given photographic situation. However, this is a challenging problem because the performance of image deblurring depends on various factors such as the scene statistics, the characteristics of applied blur, and exposure parameters used for the capture.

This dissertation provides a theoretical analysis that answers an important part of this question. In particular, a method of estimating the performance of multi-image deconvolution based on a noise amplification factor is proposed. This method relates the performance with exposure parameters, which are intuitive and easily applicable in practice. Then, this dissertation proves the following property: *Multi-image deconvolution does not benefit from increasing exposure time beyond a certain threshold*. This result goes against the common thought that longer exposure times improve the image quality. When camera shake blur exists, increasing exposure time does not help because the noise introduced by computationally removing additional blur cancels the reduced photon noise in captured images.

Finally, the performance of different capture strategies is compared. Our analysis

focuses on comparing multi-image deconvolution and align-and-average, which can be considered as a special case of multi-image deconvolution when handshake blur is negligible. Both approaches are compared in various scenarios to show that multi-image deblurring is applicable to a wider range of photographic situations compared to align-and-average. Then, guidelines are formulated to help users to determine how to capture and process images to get the best output in real-world situations.

1.4 Contributions

The first contribution of this dissertation is an implementation of gyro-based multi-image deblurring system that handles various practical issues. These issues include gyroscope drift correction, spatially-varying blur that includes off-center rotational blur, time synchronization between image data and gyroscope data, electronic rolling shutter, and handling moving objects and over-exposed regions. All these issues add complexity to the system, but they must be included in order to produce high-quality images.

The second contribution is an analysis that allows us to compare the performance of different capture strategies. Based on this analysis, following key points are demonstrated:

- Multi-image deconvolution does not benefit from increasing exposure time beyond a certain threshold.
- Multi-image deconvolution is a good capture strategy in low-light situations when align-and-average fails.

In the following chapter, a review of other related works on various subproblems is described. Then, the next two chapters will describe the implementation details of our gyro-based camera motion estimation (Chapter 3) and image deblurring system (Chapter 4). Following our analysis of the performance of multi-image deconvolution (Chapter 5), Chapter 6 will compare the performance of align-and-average and multi-image deconvolution and formulate guidelines for choosing good strategies. Finally, Chapter 7 will conclude the dissertation, suggesting some future research directions.

The gyro-based multi-image deblurring system is also discussed in a paper published at CVPR 2014 [46]. This dissertation includes the results of additional experiments beyond those reported in this paper.

Chapter 2

Related Work

Before we describe our image deblurring system, it is valuable to examine a number of well-studied problems in computational photography that are closely related to this system. Section 2.1 reviews the basic building blocks in multi-image deblurring. Then, Section 2.2 provides an overview on the issues related to using inertial sensors and compares previous works on image deblurring system based on inertial sensors. Finally, previous works on comparing capture strategies are presented in Section 2.3.

2.1 Improving Image Deblurring

We focus on two topics that are related to our multi-image deblurring: how camera shake blur is modeled and how multiple images are used to reduce motion blur.

2.1.1 Modeling Camera Shake Blur

When a camera moves during an exposure, a captured image becomes blurry as the sensor integrates the relative motion between the camera and the scene. The pose of the camera continuously changes during an exposure, and each pose gives a slightly different view of the same scene. The difference between these views causes camera shake blur.

Many blur models assume a spatially uniform blur [35, 54, 11], which means that

the shape of blur is identical across the entire image. This assumption allows the blur to be modeled with a well-known 2-D convolution between the scene and a blur kernel that represents the shape of the blur. The main advantage of this approach is that the frequency domain analysis is available to characterize the blur. In addition, the applied blur can be efficiently removed in the frequency domain. However, this uniform blur model is often violated in practice because in-plane rotation is usually a significant contributor to actual camera shake [36]. Thus, this blur model should be applied only in small local windows that satisfy the uniform blur assumption.

A common approach in modeling camera shake blur is to use geometric representation of camera motion, which inherently gives spatially-varying blur. Although we need six degrees of freedom (DOF) to describe full camera motion [30], 3-axis rotation [60] or in-plane translation and rotation [19] is commonly used as well. The 3-axis rotation model is a reasonable approximation because the rotational component is the dominant factor of camera shake. In practice, there are only small changes in the camera position during capture, and thus, the amount of blur caused by camera translation is negligible for sufficiently distant scenes. Köhler et al. [33] show that 3-DOF approximation of full camera motion agrees well with real camera shake measurements.

2.1.2 Multi-Image Approaches to Reducing Motion Blur

In addition to single-image deblurring methods, a number of multi-image approaches based on different data captures have been proposed.

One category of approaches relies on align-and-average methods. In astronomical photography, lucky imaging methods have been popularly used, dating back to the 1970s [17]. These methods select images that have undergone the least degradation due to atmospheric turbulence, and merge them after alignment. This idea is extended in various recent works in computational photography to reduce the amount of motion blur or to denoise video stream. These methods are predicated on accurate alignment between images, which can be done based on a global transformation [56], a global alignment with local adjustments [29], optical flow [39, 49], or gyroscope

measurements [51].

Another category is called multi-image deblurring. Papers taking this popular approach share the observation that combining images with different shapes of blur makes deblurring stable [50, 12, 11]. Li et al. [38] remove blur in video frames to generate a sharp panorama image. Agrawal et al. [5] deliberately vary exposure times to capture blurry images with different shapes of blur for an object moving at a constant speed. Cho et al. [13] estimate multiple motions within the scene. However, these methods are based on blind estimation, which is still a challenge. We use gyroscope measurements to make the estimation non-blind.

Other sets of data are assumed to improve the image quality in low-light situations. A pair of flash and no-flash images [48], or a pair of noisy and blurry images [63, 57] can be captured. These methods aim to fuse only the good properties of the two images. Sharp image details come from flash image or noisy image, while good color information is transferred from long exposure shot.

2.2 Inertial Sensors and Their Applications in Images and Videos

The emergence of MEMS inertial sensors enabled new approaches to solve a number of well-studied problems in computational photography and computer vision. By using the rich data provided by inertial sensors, the complexity of handling image and videos captured from hand-held devices is significantly reduced. For example, applications such as video stabilization and rolling shutter correction [31, 20, 26], motion tracking [24, 37] and camera shake removal [30, 7, 55, 51, 46] benefit from this new technology and many solutions are efficient enough to be implemented in mobile devices.

In this section, we address a number of practical issues in combining image data and inertial sensor data, and review previous works that tackle these problems. Then, we focus on image deblurring systems that use inertial sensors and compare them with our system.

Technology	Principle	Applications	Price per axis
RLG, FOG	Sagnac effect	Navigation systems	$> \$ 2000$
MEMS	Coriolis force	Consumer electronics	$< \$ 2$

Table 2.1: An overview of gyroscope technologies. Although conventional gyroscope technologies can provide higher performances, MEMS gyroscopes dominate consumer electronics market because of their low requirements in size, power and cost [6, 58].

2.2.1 Gyroscope Technologies

Before discussing details about using inertial sensors, a brief overview of gyroscope technologies is provided, which is summarized in Table 2.1. More details can be found in [34, 62].

Conventional gyroscope technologies, such as ring laser gyroscopes (RLG) and fiber optic gyroscopes (FOG), are based on the Sagnac effect. These gyroscopes monitor the difference in the propagation time between two laser beams, which is affected by the rate of sensor rotation. Because of their high performance, these gyroscopes are used in inertial navigation systems in guided missiles, submarines and aircrafts. However, their size and cost are prohibitively large to be used in consumer mobile devices.

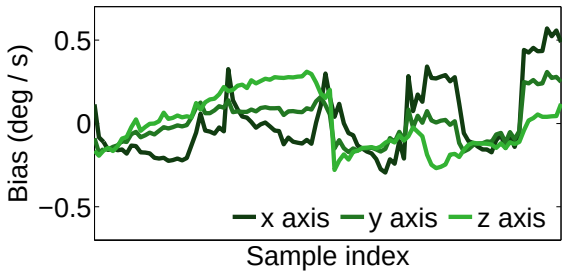
In contrast, MEMS gyroscopes can be fabricated in micrometer scales. Most MEMS gyroscopes contain a vibrating mechanical structure that is affected by the Coriolis force caused by the sensor rotation. By sensing the Coriolis force, the angular velocity of the sensor is measured. These micromachined, low-power and low-cost gyroscopes are widely adopted in consumer devices, enabling numerous novel applications in mobile platforms. However, their performance is often degraded by unstable bias, and practical issues must be properly handled to obtain more reliable measurements. The following section will describe how these challenges are tackled.

2.2.2 Challenges in Using Inertial Sensor Measurements

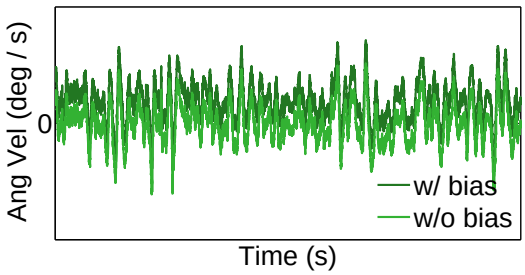
Here, we list a few practical issues that affect the accuracy and reliability of inertial sensor measurements:

- **Unstable bias** A bias refers to zero-rate output, which is a non-zero offset that exists independent of sensor motion. The amount of bias is affected by various factors, such as temperature, vibrations and mechanical stress [6, 58], and thus slowly varies with time.
- **Unknown time delay between image data and sensor data** Timestamps obtained from different sensors in the system suffer from unknown time delay caused by data transfer, buffering, and software stack delays in operating systems. This delay varies randomly.
- **Electronic rolling shutter** In contrast to a global shutter, an electronic rolling shutter scans across the frame either in vertical or horizontal direction to read out pixel values. Thus, each row or column is captured at a slightly different time.
- **Gravity offset** Accelerometers measure proper acceleration that includes gravitational acceleration, which must be subtracted to obtain the acceleration due to the motion.
- **Off-center rotations** The measurements from gyroscopes are not affected by the position of rotation center, and thus cannot distinguish off-center rotations with the same angular rate.

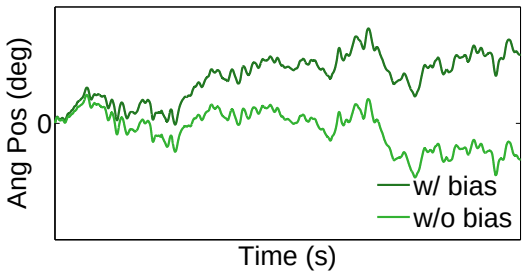
Among these issues, we discuss more about the effect of unstable bias. A bias causes drift in estimation after integration, and the amount of drift accumulates with time, as illustrated in Figure 2.1b and Figure 2.1c. When accelerometers are used, this drift becomes more critical because the measurements are integrated twice to obtain the translational component. Figure 2.1a shows the variation of bias measured from a gyroscope used in our system. Each measurement was obtained more than 30 minutes



(a) Bias measurements



(b) Bias in gyroscope data

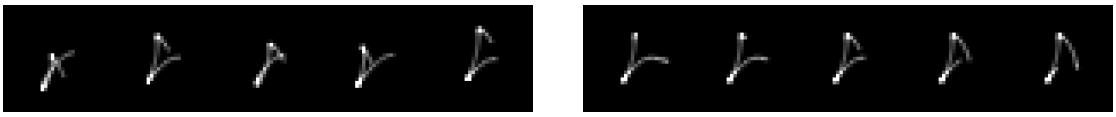


(c) Drift caused by bias

Figure 2.1: Gyroscope drift caused by unstable bias. (a) Unstable bias in a gyroscope. Each sample represents the bias measurement for a gyroscope used in our system (Invensense MPU-6050). (b) Two gyroscope data are simulated with and without applying a bias, which is modeled as a constant offset. (c) The bias causes drift after integration. The gap between two curves represents the amount of drift. Note that the bias may cause more than a pixel deviation when the time is longer than $\frac{1}{6}$s for our system.

after the previous one. The standard deviation of the samples is $(0.21, 0.12, 0.17)°/s$ for x, y and z axis, respectively, which produces more than a pixel of deviation if the total capture time is longer than $\frac{1}{6}s$ for our system. When a long period of time is considered, this bias must be properly compensated.

Figure 2.2 demonstrates how these issues may affect the result of blur kernel estimation, which is a common task in image deblurring. Different bias values make the end point of motion trajectory drift in a blur kernel as shown in Figure 2.2a. When multiple images are considered, this drift also causes misalignment between frames. Figure 2.2b shows that blur kernels may be estimated from the camera motion that corresponds to a wrong time window because of unknown time delay.



(a) Gyroscope drift (b) Gyroscope time delay

Figure 2.2: The effect of gyroscope drift and time delay on a blur kernel. A set of blur kernels is simulated with different values of gyroscope bias and time delay, while the exposure time for each blur kernel is not changed. (a) Each blur kernel is obtained by applying different bias values that are randomly selected. Gyroscope drift affects the end point position of the motion trajectory. (b) Each blur kernel reflects the motion within a time window that is shifted by a time delay. Five samples of time delays with the same spacing were selected.

Thus, most applications that capture both image frames and inertial sensor measurements must be designed to handle these practical issues. The requirement for estimation accuracy varies between different applications. For example, video stabilization may tolerate small amount of error while image deblurring usually requires a pixel-accuracy blur estimation to avoid artifacts and minimize remaining blur.

Kelly and Sukhatme [32] propose a self-calibration method that estimates the sensor-to-sensor transform, the gyroscope and accelerometer biases and the gravity vector based on the unscented Kalman filter. However, they use inertial sensors that have higher performance than the ones in current mobile devices and model sensor bias as Gaussian random walk process, which may not hold in practice. Jia and Evans [27] propose an online calibration method for cellphone camera and gyroscope based on an extended Kalman filter, and consider electronic rolling shutter and the time delay as well. Karpenko et al. [31] and Ringaby and Forssén [51] also calibrate gyroscope to handle electronic rolling shutter, gyroscope bias and time delay. However, all these methods rely on feature point tracking applied to video frames, which may become less accurate if images are degraded by handshake blur.

When both gyroscope and accelerometer are used, bias in gyroscope is often neglected because the drift caused by an accelerometer is much more significant [30, 7]. Joshi et al. [30] repeatedly apply their image deblurring method with possible bias values and then picks the best result. This approach is slow and only the bias in an accelerometer, not gyroscope, is handled. Bae et al. [7] correct accelerometer drift

by comparing the histograms of blur kernels. Our gyroscope drift correction method takes advantage of capturing multiple images, while these two methods are limited to the single-image case.

2.2.3 Reducing Camera Shake Blur Using Inertial Sensors

The most commercially successful solution for reducing camera shake blur is optical image stabilization embedded in consumer digital cameras. An optical image stabilizer shifts either a floating lens element or the image sensor to compensate for the vibration detected by gyroscopes and keeps the optical path to the sensor stable. Recently, optical stabilizers are also adopted in cell phone cameras such as the ones in Nokia Lumia 920 [1] and HTC One M7 [2]. Although these stabilizers are known to be effective, they add size and cost to traditional camera modules.

A different direction of research focuses on computational methods for reducing camera shake blur. Table 2.2 summarizes the previous approaches that use inertial sensor for camera motion estimation. Joshi et al. [30] propose a single image deblurring method that uses both gyroscope and accelerometer. This method handles 6-DOF camera motion with accelerometer drift correction. Bae et al. [7] extend the previous method to handle depth-dependent blur by using a depth sensor. Šindelář and Šroubek [55] implement a gyro-based image deblurring method in a mobile phone.

Although these previous methods show some promise, they are limited to capturing a single image. In contrast, our method benefits from capturing a burst of images. There are two main advantages in the multi-image approach. First, deconvolution becomes significantly more stable. Second, practical issues such as gyroscope drift can be handled more efficiently by using multiple images. Ringaby and Forssén [51] also capture multiple images, but the result is obtained by averaging captured images after alignment. Our method can further reduce the amount of blur and is more robust in low-light situations.

Our method uses only gyroscope data, not accelerometer data, because rotation is the dominant factor that causes blur and the translation of a camera can be ignored for sufficiently distant scenes. Another reason is that the bias in accelerometers is very

	Number of images	Inertial sensors	Gyro drift correction	Processing method
Joshi et al. [30]	single	accel, gyro	no	image deblurring
Bae et al. [7]	single	accel, gyro	no	image deblurring
Šindelář and Šroubek [55]	single	gyro	no	image deblurring
Ringaby and Forssén [51]	multiple	gyro	yes	align-and-average
Our method	multiple	gyro	yes	image deblurring

Table 2.2: Previous works that use inertial sensors for reducing camera shake blur.

strong and separating gravity from their measurement is not trivial. Instead, we correct the drift in gyroscope data and handle off-center rotations without accelerometer data to make an accurate estimation of rotational movements.

2.3 Finding Good Capture Strategies

Capturing a good set of data is a key task that can improve the performance of computational photography applications. There exist many previous works that focus on what capture strategies are best for their applications.

Several papers on high dynamic range (HDR) imaging address this issue of finding a set of optimal parameters. Hasinoff et al. [22] show that the optimal choice of exposure time and analog gain significantly improves the signal-to-noise ratio (SNR) of the merged HDR result. This improvement comes from using high analog gains as they are useful for suppressing read noise while reducing the total capture time. This idea is adopted in a few HDR metering methods that determine scene-specific parameters in real time on mobile devices [53, 18].

Comparing the capture strategies for camera motion deblurring has been studied by Zhang et al. [64]. Their analysis concludes that align-and-average works better than single-image deconvolution when total capture time is kept the same. Mitra et al. [43] prove that deblurring a long exposure shot is better than taking a short exposure shot in low-light situations. However, their blur model is limited to 1-D

rectilinear blur. Our analysis, for the first time, extends previous analyses to the case when multi-image deconvolution is used. In addition, our analysis is based on real measurements of camera shake blur and is applicable to more practical situations.

Boracchi and Foi [8] perform a series of experiments to show the relation between the performance of single-image deconvolution and exposure time and observe that restoration error of single-image deconvolution is stabilized after reaching a minimum. We prove this observation for the case of multi-image deconvolution and suggest how capture parameters can be selected based on this observation.

Chapter 3

Gyro-Based Camera Motion Estimation

Any image deblurring method requires solving two subproblems: camera motion estimation and non-blind image deblurring. When gyroscope data is available, these two subproblems significantly benefit from direct measurement of camera motion. In this chapter, we describe our approach to the first subproblem: gyro-based camera motion estimation.

To start with, we present our image formation model that describes the characteristics of noise and camera shake blur, which is an essential component of any theoretical analysis of image deblurring method. Then, our gyroscope drift correction method is proposed to tackle various practical issues encountered when gyroscope data is used along with image data.

3.1 Image Formation with Camera Motion

The principal drawback of using a short exposure time in low light is image noise. Lengthening the exposure decreases noise, while it increases handshake blur. It is therefore important that we define a noise model that allows us to handle handshake blur. We begin by describing an image noise model for when the camera is stable, and then extend it to include the effect of camera motion.

3.1.1 Image Noise Model

We adopt a common affine model [40, 22] to describe the noise characteristics of a captured image and show how the characteristics depend on the scene and capture parameters. This model assumes that image noise consists of two components: photon noise and read noise. Depending on the photographic situation, the dominant component of these two types of noise determines the characteristics of total image noise.

Photon noise reflects the fluctuation in the number of photons that are collected by a pixel. The total number of collected photons, which follows the Poisson distribution, is determined by the scene brightness and the duration of exposure. Because the variance of this noise is equal to the number of collected photons, photon noise becomes more important when the scene is brighter and longer exposure times are used. When a large number of photons is involved, the Poisson distribution can be approximated by the Gaussian distribution whose variance is related as

$$\sigma^2_{photon} \propto t\Phi \tag{3.1}$$

where σ^2_{photon} is the photon noise variance, t is exposure time and Φ is scene brightness.

In contrast, read noise includes other noise sources that are scene-independent, such as the noise coming from the image sensor readout circuitry and quantization noise. Figure 3.1 shows our image noise formation model. Two Gaussian noise sources are assumed before and after analog amplification. This comes from the observation that the properties of read noise are also affected by the choice of analog gain [64, 22]. The variance of read noise σ^2_{read} is represented as

$$\sigma^2_{read} = g^2\sigma^2_{r_0} + \sigma^2_{r_1} \tag{3.2}$$

where g is analog gain, $\sigma^2_{r_0}$ and $\sigma^2_{r_1}$ are the variances of readout noise applied before and after analog amplification, respectively. This two-component model is valid when exposure times are short enough to avoid dark current noise, typically much less than a second.

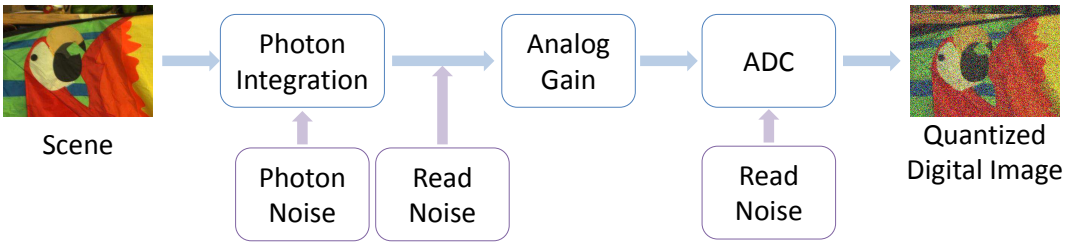


Figure 3.1: Our image noise model. We assume images captured by an image sensor are degraded by two types of noise: photon noise and read noise. Photon noise reflects the fluctuation in the number of collected photons, while read noise represents scene-independent noise sources that are applied before and after analog amplification.

We can formulate the noise model in terms of SNR. The affine imaging model describes raw pixel values of image Y as the sum of noise-free image X and the zero-mean Gaussian noise N with standard deviation σ_N:

$$Y = X + N. \tag{3.3}$$

The variance of N, σ_N^2, is modeled as the sum of photon noise variance σ_{photon}^2 and read noise variance σ_{read}^2. The SNR of a captured image Y is written as

$$\mathrm{SNR}(Y) = \frac{X}{\sigma_N} = \frac{X}{\sqrt{\sigma_{photon}^2 + \sigma_{read}^2}} = \frac{c_1 g t \Phi}{\sqrt{c_2 g^2 t \Phi + g^2 \sigma_{r_0}^2 + \sigma_{r_1}^2}}. \tag{3.4}$$

where c_1 and c_2 are conversion factors. The raw signal value is based on the assumption that the camera has a linear response, which usually holds for raw image data.

When a very short exposure time is used, read noise is the dominant factor, i.e., $\sigma_{read}^2 \gg \sigma_{photon}^2$. In this case, increasing exposure time improves the SNR linearly:

$$\mathrm{SNR}(Y) \propto t. \tag{3.5}$$

When exposure time becomes longer than a certain threshold, photon noise becomes

dominant, and the SNR increases as the square root of exposure time:

$$\mathrm{SNR}(Y) \propto \sqrt{t}. \tag{3.6}$$

This threshold shifts depending on the scene brightness. For darker scenes, this transition occurs in longer exposures.

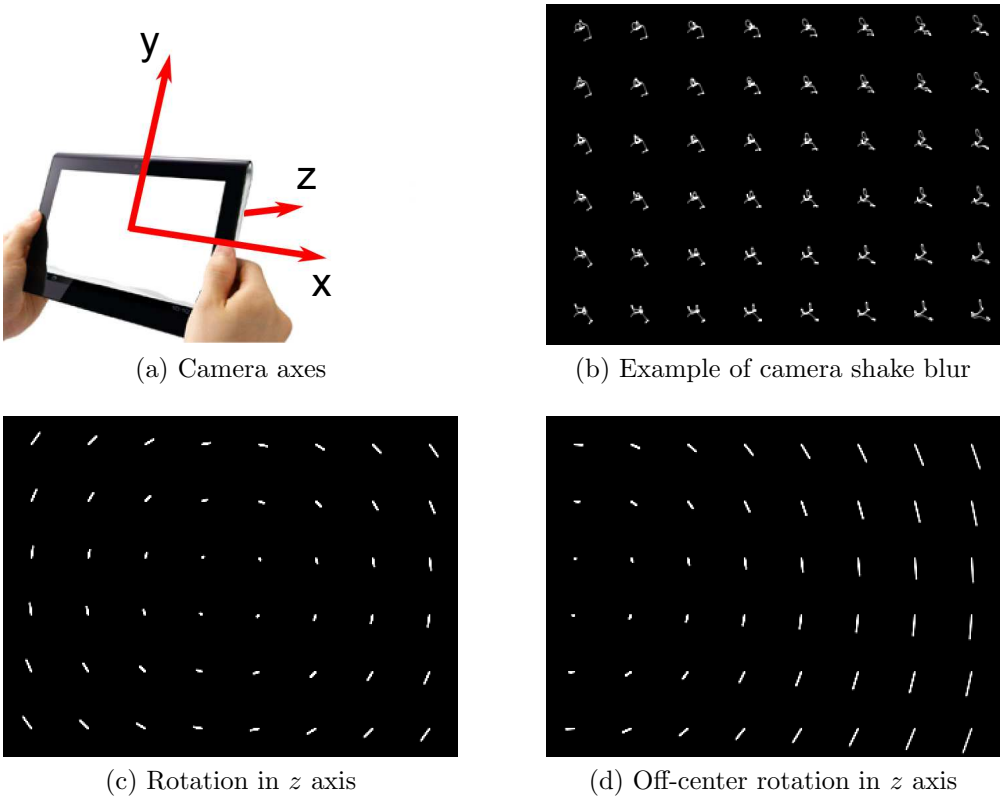
3.1.2 Modeling Spatially-Varying Camera Shake Blur

We use a geometric model to represent spatially-varying blur, and then adopt a uniform blur model as a local approximation. The geometric model expresses the blur as the sum of global transformations that represent a series of camera rotations. We assume that camera motion is dominated by rotation. This is a common assumption in image stabilization systems and is valid for sufficiently distant scenes. The entire image is then divided into small rectangular blocks that are assumed to contain the spatially uniform blur. This approximation allows us to apply a theoretical analysis that provides useful insights into the deblurring process while reducing the complexity of implementation. Our model assumes that the scene is static and does not directly handle local motions. In addition, our model does not account for other types of blur, such as depth-dependent defocus blur and optical blur.

Figure 3.2 shows examples of camera shake blur represented by our model. The camera axes are defined as shown in Figure 3.2a, and Figure 3.2b shows an example of real blur. Like some other geometric models [60], our geometric model represents the angular rotation of the camera. However, the center of rotation is modeled as well because gyroscope measurements are not affected by the rotation center. Note that two camera motion shown in Figure 3.2c and Figure 3.2d are not distinguished with gyroscope measurements. We represent the off-axis angular rotation at point $\boldsymbol{r}$ as

$$R_{\boldsymbol{o}}^{\Delta\boldsymbol{\theta}}\{\boldsymbol{r}\} = R^{\Delta\boldsymbol{\theta}}(\boldsymbol{r} - \boldsymbol{o}) + \boldsymbol{o} \tag{3.7}$$

where $R^{\Delta\boldsymbol{\theta}}$ is a 3-axis rotation matrix for angular rotation $\Delta\boldsymbol{\theta}$, and $\boldsymbol{o} = (o_x, o_y, o_z)^T$ is



(a) Camera axes

(b) Example of camera shake blur

(c) Rotation in z axis

(d) Off-center rotation in z axis

Figure 3.2: Our model for representing camera shake blur. We use a geometric model to represent spatially-varying blur based on 3-axis rotation of the camera. Subfigure (a) shows how the camera axes are defined. An example of camera shake blur represented by our model is shown in (b). The center of rotation is also modeled because gyroscope measurements cannot distinguish two rotations shown in (c) and (d).

the rotation center. If a rotation $\Delta\boldsymbol{\theta}(t_s, t_t)$ is applied during time window $[t_s, t_t]$, then jth pixel $\boldsymbol{p}^j = (x^j, y^j)^T$ in the image plane is shifted to $(x^{j\prime}, y^{j\prime})^T$ by the following relation defined by $H_{\Delta\boldsymbol{\theta}(t_s,t_t)}$ in homogeneous coordinates:

$$\alpha(x^{j\prime}, y^{j\prime}, 1)^T = H_{\Delta\boldsymbol{\theta}(t_s,t_t)}(x^j, y^j, 1)^T \tag{3.8}$$

$$= M_f R_{\boldsymbol{o}}^{\Delta\boldsymbol{\theta}(t_s,t_t)} \{M_f^{-1}(x^j, y^j, 1)^T\} \tag{3.9}$$

where α is an unknown scaling factor, and M_f is the camera intrinsic matrix, which depends on focal length f. Given this model for rotation, the rotation-blurred image Y

can be formulated as the sum of the time integration of all geometrically transformed latent images during the exposure window $[t_s, t_t]$ plus noise:

$$Y = \int_{t_s}^{t_t} A_{\Delta\boldsymbol{\theta}(t_s,t)} X dt + N \tag{3.10}$$

where $A_{\Delta\boldsymbol{\theta}}$ represents the image resampling operator that performs the transformation $H_{\Delta\boldsymbol{\theta}}$ to all $\boldsymbol{p}^j$, X is the latent sharp image, and N is the Gaussian noise whose variance is given as in Equation 3.4. Note that $H_{\Delta\boldsymbol{\theta}}$ defines spatially-varying blur.

The blur model expressed in Equation 3.10 can then be approximated with the uniform blur model for a small local window. A local window of blurry image Y_i around $\boldsymbol{p}^j$, which is assumed to contain uniform blur, is represented as

$$Y_i = K_i^j \otimes X + N_i \tag{3.11}$$

where K_i^j is the 2-D blur kernel, $\otimes$ is the 2-D convolution operator, and i denotes the input image index when multiple images are available. The kernel K_i^j is specified by the exposure time window $[t_s^i, t_t^i]$, which can be expressed in terms of capture parameters. The two ends of the window are defined as follows:

$$t_s^i = t_s^1 + (i-1)t_{ft} \tag{3.12}$$

$$t_t^i = t_s^i + t_{exp} \tag{3.13}$$

where t_{exp} is the exposure time of a single shot and the frame time t_{ft} is the interval between frames. Then, the blur kernel K_i^j is expressed as

$$K_i^j = \frac{1}{n_k} \sum_{k=0}^{n_k-1} \delta(\boldsymbol{p}^j_{t_s^i + k\Delta t} - \boldsymbol{p}^j_{t_s^1}) \tag{3.14}$$

where the time window is discretized with n_k samples and the interval $\Delta t = \frac{t_t^i - t_s^i}{n_k - 1}$, and $\delta(\cdot)$ is the discrete unit impulse function. Note that all entries in K_i^j sum to one. In addition, t_s^1, the time when the first exposure starts, is set as the reference time so that relative rotations between input images are preserved in the kernel.

3.2 Measuring Camera Motion Using a Gyroscope

Although gyroscope measurements can be directly applied to evaluate the angular rotation in Equation 3.8, a drift correction method is applied beforehand whenever capture takes place to tackle the practical issues described in Section 2.2.2. Our drift correction method finds the best camera motion estimation that correlates gyroscope measurements with captured image frames.

We model the angular rotation by accumulating the incremental changes given as

$$\Delta\boldsymbol{\theta}(t, t+\Delta t) = (\boldsymbol{\omega}(t+t_d) + \boldsymbol{\omega}_d)\Delta t \tag{3.15}$$

where $\boldsymbol{\omega}(t)$ is the rate of rotation measured with gyroscope, t_d is the time delay, $\boldsymbol{\omega}_d$ is the bias which causes drift, and Δt denotes the time interval between gyroscope samples. Our method estimates the time delay t_d, the bias $\boldsymbol{\omega}_d$ and the rotation center $\boldsymbol{o} = (o_x, o_y, o_z)^T$. We assume that the rotation center $\boldsymbol{o}$ does not move in z direction, $o_z = 0$, and the variation in $\boldsymbol{o}$ is negligible during the capture.

Our drift correction method is based on a series of pairwise comparisons and takes advantage of capturing multiple images. The pairwise comparison, which is demonstrated in Figure 3.3, quantifies how well the captured images agree with the estimated camera motion. Similar cost functions are proposed in visual tracking [28] and image deblurring techniques [50]. Let us assume two capture images, which are blurry, are denoted as Y_1 and Y_2. Their blur kernels, represented as K_1 and K_2, respectively, are obtained based on gyroscope measurements and current estimation of $\boldsymbol{\omega}_d$, t_d and $\boldsymbol{o}$. Then, the two images are blurred again with the other image's blur kernel. These two re-blurred images should be the same in the ideal case because the order of applying blur should not matter, i.e., convolution operator is commutative. The aggregated error between the two images represents the match score of current estimation of camera motion.

Here, our complete drift correction method is described. We search for six parameters, $\boldsymbol{\omega}_d$, t_d, o_x and o_y, which minimize the objective function.

First, we find a reference input image $Y_{i_{ref}}$ that is observed to be the sharpest among the n input images. We assume that the sharpest image is the one whose blur

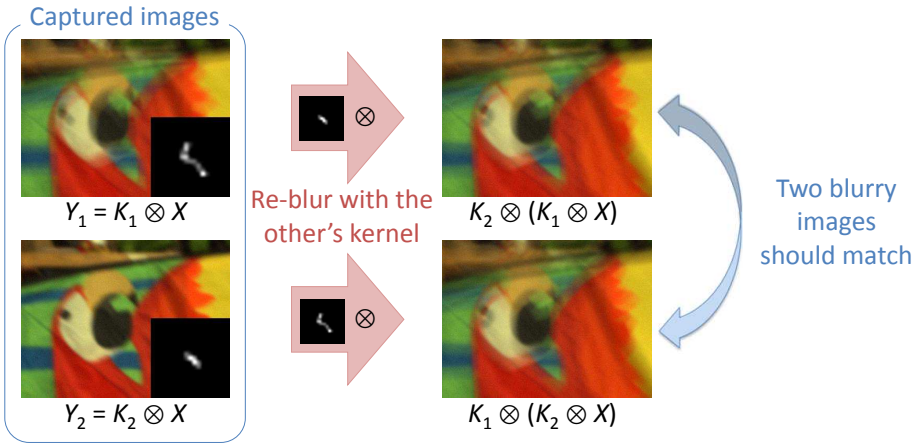


Figure 3.3: A pair-wise comparison for gyroscope drift correction. This pairwise comparison quantifies how well the captured images agree with the estimated camera motion.

kernel has maximum spatial variance.

Second, we find n_c image blocks that contain sharp corners in $Y_{i_{ref}}$. Rather than considering the entire scene, only a few number of distinct image blocks are used to reduce computational cost. These image blocks are assumed to have strong corner response [21] and contain uniform blur. The centers of the blocks are denoted as $\{\boldsymbol{c}_j\}_{j=1,\cdots,n_c}$. We typically select $n_c = 12$ and the size of an image block to be equal to the blur kernel size.

Finally, we formulate a minimization problem:

$$\underset{\boldsymbol{\omega}_d, o_x, o_y, t_d}{\operatorname{argmin}} \sum_{i \neq i_{ref}} \sum_{j} \|W_i^j (K_{i_{ref}}^j \otimes Y_i - K_i^j \otimes Y_{i_{ref}})\|^2 \tag{3.16}$$

where W_i^j is the weight applied to the jth block of the ith image. Note that this objective function is evaluated by performing $(n-1)$ pairwise comparisons with respect to the reference image $Y_{i_{ref}}$. The weight is set as $W_i^j = w_j w_i W^j W_i$ where W^j is the binary mask that selects the local window around $\boldsymbol{c}_j$, W_i reflects the magnitude of gradient in Y_i, w_j is the distance from $\boldsymbol{c}_j$ to the image center, and $w_i = |i - i_{ref}|$ is the temporal distance from the sharpest input image. The role of W_i is to avoid flat regions that only contain noise. In addition, w_j and w_i are introduced to weight the

region that is more affected by the drift and to give consistent estimation throughout the camera motion.

The effect of electronic rolling shutter is considered when K_i^j is obtained. Without loss of generality, let us assume electronic rolling shutter scans across the sensor vertically, from top to bottom. When blur kernels are generated by Equation 3.14, the exposure time window $[t_s^i, t_t^i]$ is shifted as

$$[t_s^i, t_t^i] + \frac{j_y}{n_y} t_{rs} \tag{3.17}$$

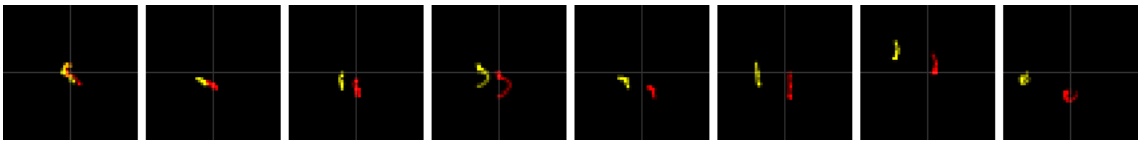
where t_{rs} is the time required to readout whole image, j_y is the image row of pixel j and n_y is the image height. Our image sensor has $t_{rs} = \frac{1}{15}s$.

This optimization is performed using coordinate descent. To find a minimum of the objective function, line search is performed for each variable in each iteration. Each variable takes steps until the objective function is reduced, and the search direction is reversed with a reduced step size if no improvement is made. This optimization terminates when all variables reach a target precision and a local minimum is found. First, we search for $(\boldsymbol{\omega}_d, o_x, o_y)$ in a multi-scale pyramid to avoid prematurely accepting local minima. Then, t_d is estimated at the finest scale, followed by refining other parameters based on new t_d. Each step usually converges quickly, within 30 iterations.

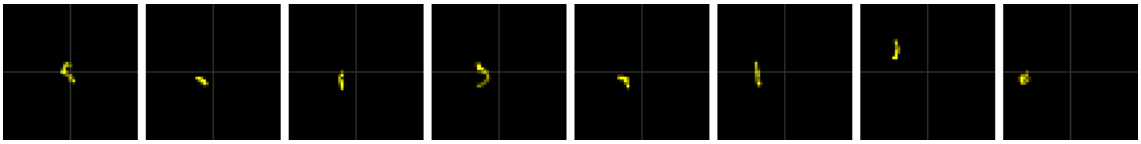
Figure 3.4 shows the effectiveness of our drift correction. A burst of eight images is captured to generate the output image in Figure 4.3b, and an inset is shown here. Figure 3.4a shows two sets of blur kernels obtained before and after applying our drift correction in red and yellow, respectively. The gap between these two sets shows the amount of drift that is compensated by our method. This correction makes significant improvement in the deblurred result, as shown in Figure 3.4d and Figure 3.4f. An additional experiment is done to demonstrate the effect of rolling shutter correction. The blur kernels after the correction are repeated in Figure 3.4b, and Figure 3.4c is obtained by applying our drift correction with $t_{rs} = 0$ to neglect the effect of rolling shutter. Note that the shape of blur kernels is slightly different between the two cases. This difference makes the result in Figure 3.4e more blurry than the result with all

corrections in Figure 3.4f.

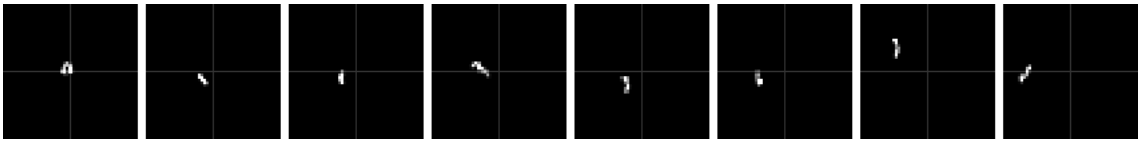
Compared to other single-image approaches [30], our method is efficient because the latent image is not necessary for the correction. Recovering the latent image requires complicated operations such as image deblurring and the latent image often contains deblurring artifacts.



(a) Blur kernels before and after drift correction



(b) Drift correction with rolling shutter correction



(c) Drift correction without rolling shutter correction



(d) Before correction (e) w/o rolling shutter correction (f) With all corrections

Figure 3.4: The effectiveness of our gyroscope drift correction. Eight images are captured to generate the output image shown in Figure 4.3b. Subfigure (a) shows two sets of blur kernels before and after drift correction, which are represented in red and yellow, respectively. The deviation between the red and yellow kernels shows that the drift is actually significant. When image deblurring is applied, the output image in (f) shows clear improvement after the correction. Two sets of blur kernels presented in (b) and (c) shows the difference caused by rolling shutter effect. When the rolling shutter correction is not handled, the result image becomes slightly blurry as shown in (e).

Chapter 4

Gyro-Based Multi-Image Deblurring

Combining our gyro-based camera motion estimation and non-blind image deblurring together, we have built a complete system that produces high-quality images in low-light situations. Figure 4.1 shows an overview of our gyro-based multi-image deblurring system. We use a tablet to capture a burst of images while gyroscope data is recorded simultaneously. Then, our software processes these data to make an accurate estimation of camera motion by handling practical issues including gyroscope drift, time synchronization between image data and gyroscope data, electronic rolling shutter and rotation center estimation as described in Chapter 3. Finally, an output image is recovered via non-blind multi-image deconvolution.

In this chapter, we detail our implementation of image deblurring system and provide real example images obtained from our system along with a few failure cases. These failure cases are caused by moving objects and over-exposed regions in the scene, which do not follow our image formation model. We demonstrate how the artifacts caused by these regions can be reduced by introducing an additional image blending operation.

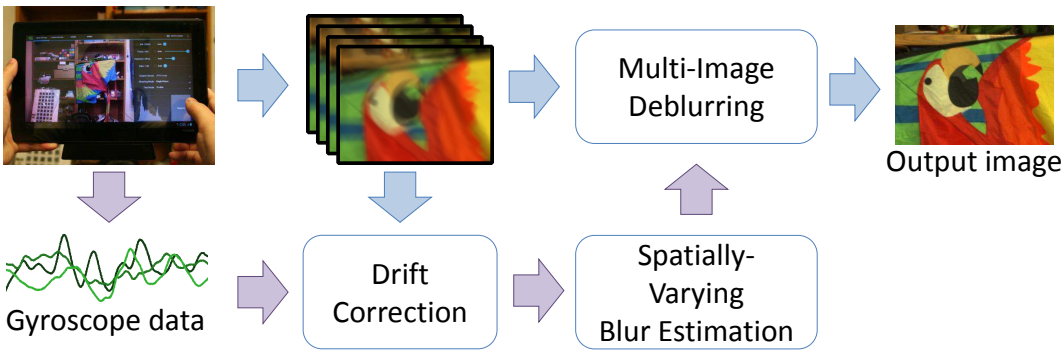
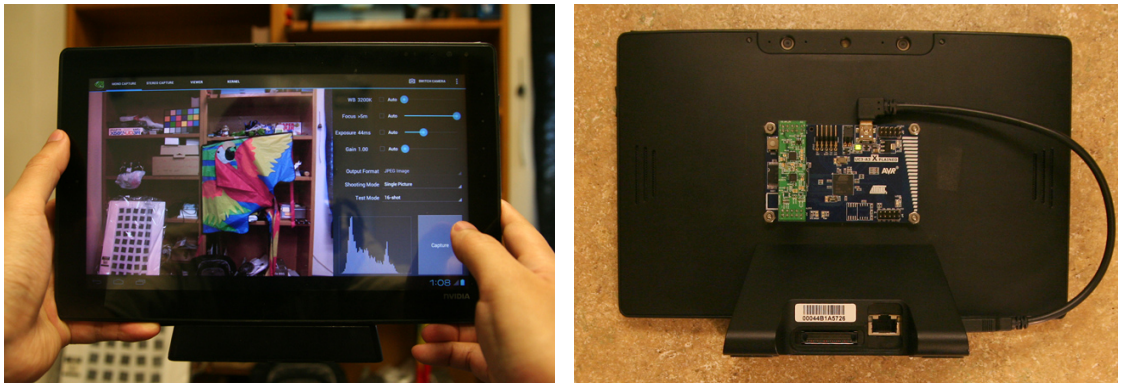


Figure 4.1: An overview of our gyro-based multi-image deblurring system.

4.1 Hardware Platform

Our capture application is implemented on an NVIDIA Tegra 3 Android developer tablet that supports FCam API [4] for flexible capture configurations, as shown in Figure 4.2. The tablet is modified to rigidly attach an Atmel UC3-A3 Xplained and Invensense MPU-6050 [25] sensor board to obtain unfiltered raw sensor data at high data rate. The sensor data is sent to the tablet through a USB connection at a maximum rate of 750 Hz. Human hands typically shake with a frequency of 10 Hz to 20 Hz [52], so this is more than enough for our purposes. The camera is able to capture 5M-pixel raw data at a maximum rate of 4 fps, and a maximum of $n = 16$ images can be captured with a burst mode.

Before the tablet is used in experiments, a one-time calibration is performed to find parameters of the image formation model. A standard geometric calibration procedure [9] is applied to find the camera intrinsic matrix M_f in Equation 3.9 and to remove geometric distortions. Note that the focal length of the camera f is included in M_f. In addition, the characteristics of the image sensor are estimated from a photometric calibration [23, 53]. First, a set of dark frames, which are images captured with the lens completely covered, is captured with different analog gains to find the read noise variance $\sigma_{r_0}^2$ and $\sigma_{r_1}^2$. Also, the mean of the dark frames is estimated for each analog gain factor so that it is subtracted from all raw data captures to compensate for the non-zero black level. Second, we capture another set of images that are lit by uniform illumination with different brightness levels. The brightness Φ is measure



(a) Front (b) Back

Figure 4.2: A tablet with a gyroscope sensor board attached. An NVIDIA Tegra 3 Android developer tablet is modified to attach an Invensense MPU-6050 sensor board to obtain unfiltered raw sensor data. The sensor data is sent to the tablet through a USB connection.

by a light meter, and the conversion factors c_1 and c_2 in Equation 3.4 are estimated by the least-squares method. For this estimation, we use the pixel values from the region around the optical center because they suffer less from the vignetting. Finally, the duration of rolling shutter t_{rs} is estimated. We follow a calibration process that is similar to the method described in [31]. After performing these calibrations, raw data captured from the image sensor can be properly processed based on the calibration result.

4.2 Removing Camera Shake Blur

Our system provides two options for reducing handshake blur: multi-image deblurring and align-and-average. These methods are applied to the same set of captured data. Note that they also use the same estimation result of the camera motion measured by a gyroscope.

4.2.1 Multi-Image Deblurring

Our non-blind multi-image deblurring is implemented by applying multi-image deconvolution to small image blocks. We assume the camera shake blur contained in each image block can be well approximated by uniform blur, and this assumption usually holds in practice for the block size we tested. The scene is divided into 36×24 blocks where each block covers about 2° of the field of view. An additional margin is added based on the blur size to avoid boundary artifacts.

Our multi-image deconvolution is done by minimizing the objective function:

$$\sum_{i=1}^{n} \|Y_i - K_i \otimes X\|^2 + \lambda \|\nabla X\|^\alpha \tag{4.1}$$

where λ is the regularization weight and ∇X represents the gradients of X. This objective function consists of two terms: the data term and the regularization term. The data term denotes the sum of mean square error (MSE) between input images Y_i and the modeled image. The regularization term represents the image prior model that is assumed for the deconvolution and the exponent α in Equation 4.1 determines the model. Here, we consider two models that are implemented in our system: the Gaussian prior and the sparse prior.

The Gaussian prior corresponds to the case when the exponent $\alpha = 2$. The main advantage of this model is that a simple analytic solution is available. In the frequency domain, the optimal estimate $\bar{X}$ is derived as

$$\bar{\mathcal{X}} = \frac{\sum_{i=1}^{n} \mathcal{K}_i^* \mathcal{Y}_i}{\sum_{k=1}^{n} |\mathcal{K}_k|^2 + \lambda(|\mathcal{G}_x|^2 + |\mathcal{G}_y|^2)} \tag{4.2}$$

where G_x and G_y are the gradient filters in x and y directions, respectively, $\bar{\mathcal{X}}$, $\mathcal{K}_i$, $\mathcal{Y}_i$, $\mathcal{G}_x$ and $\mathcal{G}_y$ denote the Fourier transforms of $\bar{X}$, K_i, Y_i, G_x and G_y, respectively. The deconvolution can be efficiently implemented by applying the frequency domain division in Equation 4.2. Note that this derivation is equivalent to the Wiener deconvolution that will be the basis of our theoretical analysis in Chapter 5.

The sparse prior is assumed when $\alpha = 0.8$. This prior prefers sharp image edges

and thus usually gives more satisfactory results. However, the optimization is no longer convex and a closed-form solution is not available. We use an iterative re-weighted least squares process [42, 35] for this optimization and this method requires significantly more computations compared to the case when the Gaussian prior is used.

The next step requires determining the weight λ in Equation 4.1. Any deconvolution method exhibits a trade-off between reducing noise level and recovering a sharp latent image, and this trade-off is largely controlled by the weight assigned to the prior. When we have a good observation of scene that can be considered as a ground truth, for example, when we simulate or have a tripod shot available, we search for the parameter λ in the range $[0.001, 1]$ which gives the smallest MSE with the ground truth. Otherwise, we manually select one or use the default value of 0.01.

4.2.2 Align-and-Average

Because the camera motion is already estimated by using gyroscope measurements, image alignment can be accurately done without referring to image data. Let us assume n input images $\{Y_i\}_{i=1,\cdots,n}$ need to be aligned to the reference camera position, which is the position at the start time of the first exposure t_s^1, and Y_i is captured during the exposure time window $[t_s^i, t_t^i]$. If we can select a good reference time $t_{align}^i \in [t_s^i, t_t^i]$ for each i, then Y_i can be transformed to the reference by applying the angular rotation $\Delta\boldsymbol{\theta}(t_{align}^i, t_s^1)$ as in Equation 3.8.

However, in our situation, the input images may contain some amount of blur that is not negligible and image alignment between blurry images is not well defined. Here, to give an aligned image that is sharp, we propose to select $\{t_{align}^i\}_{i=1,\cdots,n}$, the time in each exposure where the camera motion is stable, as

$$t_{align}^i = \underset{t:t_s^i \le t \le t_t^i}{\operatorname{argmax}} \sum_j K_i^j(\text{Round}\{\mathbf{p}_t^j - \mathbf{p}_{t_s^i}^j\}) \tag{4.3}$$

where K_i^j is the blur kernel that represents the trajectory of the jth pixel $\mathbf{p}_t^j$ during the ith exposure, $t_s^i \le t \le t_t^i$, and Round$\{\cdot\}$ is the rounding operator. The rounding



(a) *Tequila table* (b) *Anthropologie*

Figure 4.3: Two deblurred images obtained from our system. Eight images are captured from a single burst at 5M-pixel resolution, while gyroscope data is recorded simultaneously. All eight images are jointly deblurred to obtain an output image. Two insets for each example, represented in yellow boxes, are shown in Figure 4.4 to compare different approaches.

operation is used because K_i^j is defined in the discretized coordinates.

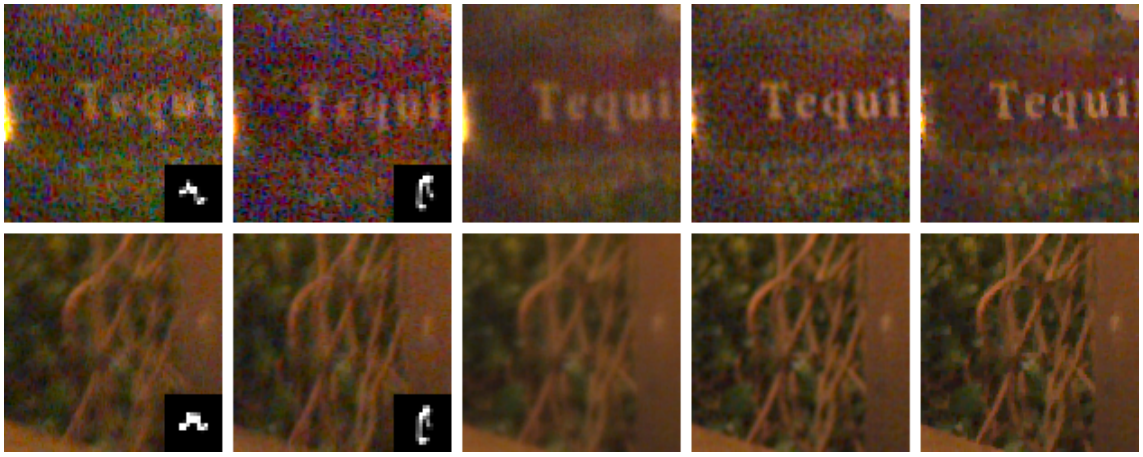
4.3 Experimental Results

Now we show deblurred images obtained from our system in Figure 4.3. A burst of eight images are captured at 5M-pixel resolution for both examples. These examples are captured with per-shot exposure time of 353ms and 177ms respectively. Two input images with typical amount of blur are selected from eight images and shown in Figure 4.4. Note that these images are noisy with some amount of blur. Our deblurred results show clear improvement over any input image, with less blur and noise, and we recover more details compared to align-and-average, with negligible artifacts. Adopting the sparse prior gives more improvement compared to the case when the Gaussian prior is used.

Figure 4.5 shows an example when there exists local motion in the scene. A burst of 16 images are captured at 5M-pixel resolution with per-shot exposure time of 135ms (Figure 4.5a). Two neutral density filters, 8× and 4×, are used to prevent

pixel saturation. An output image in Figure 4.5c is obtained by applying multi-image deblurring with the Gaussian prior. Note that the surface of water is smoothly blurred as if a long exposure shot is captured with a tripod, while handshake blur on the cliff and rocks is removed. Because image blocks with sharp details are used for our drift correction method, only the static regions are used to properly handle the drift. The errors caused by moving parts of the scene are averaged and blurred out in the output image.

The running time for generating a deblurred image in Figure 4.3b is about 24.5s for the Gaussian prior with our unoptimized single-core CPU implementation. Drift correction takes 10.5s while the rest of the time is used for deconvolution. When the sparse prior is used, running time is about 20 minutes.



Two typical input images | Align-and-average | Gaussian prior | Sparse prior

(a) Insets of *Tequila table*



Two typical input images | Align-and-average | Gaussian prior | Sparse prior

(b) Insets of *Anthropologie*

Figure 4.4: Comparison between output images obtained from different approaches. The insets of two input images are shown with their blur kernels magnified at 2× size. Note the difference in shape of blur in two insets, which demonstrates spatially-varying blur. The input images are jointly deblurred with two different priors: the Gaussian prior and the sparse prior. These two deblurred images remove handshake blur more effectively than the case when the input images are aligned and averaged. Note that the sparse prior effectively recovers sharp edges.



(a) *Bonny Dune*



(b) Input image



(c) Deblurred result

Figure 4.5: An example with local motion in the scene. Our multi-image deblurring is applied with the Gaussian prior to a burst of 16 images. Two insets represented in yellow boxes are magnified and shown in (b) and (c). Note that the surface of water is smoothly blurred as if a long exposure shot is captured with a tripod, while handshake blur on the cliff and rocks is removed.

4.4 Handling Moving Objects and Over-Exposed Regions

Although our multi-image deconvolution method effectively removes handshake blur in static regions, moving objects and over-exposed regions are not handled properly. These regions are considered as *outliers* because they do not follow our image formation model described in Equation 3.11. Moving objects may suffer from excessive blur or ghosting artifacts caused by the object motion occurred during the entire capture time. In addition, pixel values in highlights may be clipped due to limited dynamic range of image sensor, and often cause artifacts after deconvolution.

In this section, we propose an additional image blending step that follows non-blind multi-image deconvolution [47]. Our goal is not to handle outliers in a physically correct manner. Instead, we seek only to hide possible deconvolution artifacts. Figure 4.6 provides an overview of our method. First, outliers are detected in the image obtained from multi-image deconvolution. Then, the pixel values around outliers are blended with the result of a patch-based denoising method. As a result, the blended image contains sharp details in static regions, while moving objects and over-exposed regions are reconstructed naturally.

The main advantage of using a result from patch-based denoising methods is that outliers do not cause severe artifacts after denoising. These methods merge similar image patches to effectively reduce noise in a reference image [10][14]. Because of the patch search stage, they are inherently robust to moving objects or misalignment between images. After the denoising, moving objects remain as blurry as in the reference image. However, the amount of the motion blur is much less than the case when multi-image deconvolution is used. Also, over-exposed regions do not cause artifacts because they are denoised in the same way as other regions.

Let us describe more details about our method. The result image is denoted as $\widetilde{X}$, which is obtained by combining the result of deconvolution $\widetilde{X}_{dc}$ and the denoised image $\widetilde{X}_{dn}$ as follows:

$$\widetilde{X} = W\widetilde{X}_{dn} + (1 - W)\widetilde{X}_{dc} \tag{4.4}$$

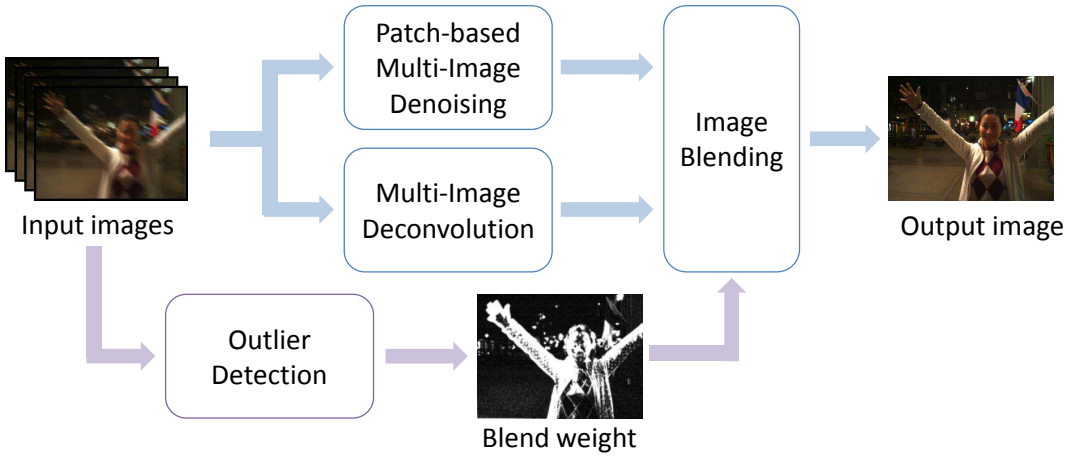


Figure 4.6: An overview of our image blending method for handling moving objects and over-exposed regions. Two images obtained from our multi-image deconvolution and a patch-based multi-image denoising method are blended to hide possible artifacts.

where W is the nonnegative blend weight that represents the result of outlier detection. The weight W is expected to have higher values on outliers to make them rely more on $\widetilde{X}_{dn}$. This is estimated by finding the maximum residual error obtained by pair-wise comparisons that are similar to the one used for gyroscope drift correction in Section 3.2. First, we select a reference image $Y_{i_{ref}}$, which is the sharpest among input images. Then, the residual error between $Y_{i_{ref}}$ and the other input images Y_i is evaluated to quantify how these images deviate from the convolution blur model. A map of residual error R_{err} is defined as

$$R_{err} = \max_{i \neq i_{ref}, c \in C} \| K_i^T \otimes K_{i_{ref}}^T \otimes (K_i \otimes Y_{i_{ref}}^c - K_{i_{ref}} \otimes Y_i^c) \|^2 \tag{4.5}$$

where $C = \{R, G, B\}$ represents color channels and Y_i^c is a color channel of Y_i. Note that R_{err} represents the maximum error in any color channel at every pixel, and K_i^T and $K_{i_{ref}}^T$ are used to align each error map to $\widetilde{X}_{dc}$. Finally, the blend weight is obtained as $W = \min(1, \beta R_{err})$ where β is a positive scaling factor that tunes the blending.

The denoised image $\widetilde{X}_{dn}$ is obtained by applying a patch-based multi-image denoising. First, all input images are globally aligned with $Y_{i_{ref}}$ by means of the affine

transform. Then, the non-local means method [10] implemented with the permutohedral lattice [3] is applied to n aligned images to obtain the result. The position vector in Gaussian filtering has six principal component analysis (PCA) dimensions for the space of image patches, in addition to two spatial dimensions that represent x and y coordinates. A patch size of 13×13 pixels is used in our implementation.

Figure 4.7a shows two examples obtained from our image blending operation. A burst of eight images was captured at 5M-pixel resolution and the total capture time was 2s. Figure 4.7b shows that outliers, a person moved during the capture and bright street lights, have high values in the weight maps to hide deconvolution artifacts. As shown in Figure 4.8d, the blended output contains only the better parts from the denoised image in Figure 4.8b and the deblurred image in Figure 4.8c, and shows minimum artifacts while recovering sharp details on the background. This image blending operation makes multi-image deconvolution more robust to handle real-world photographic situations.

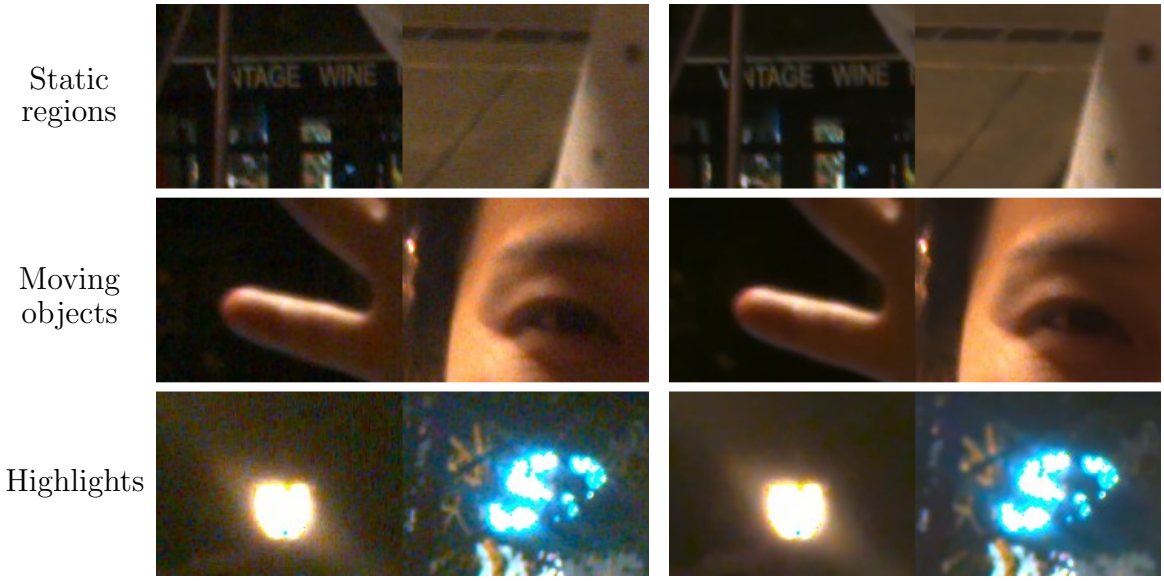


(a) Two output images

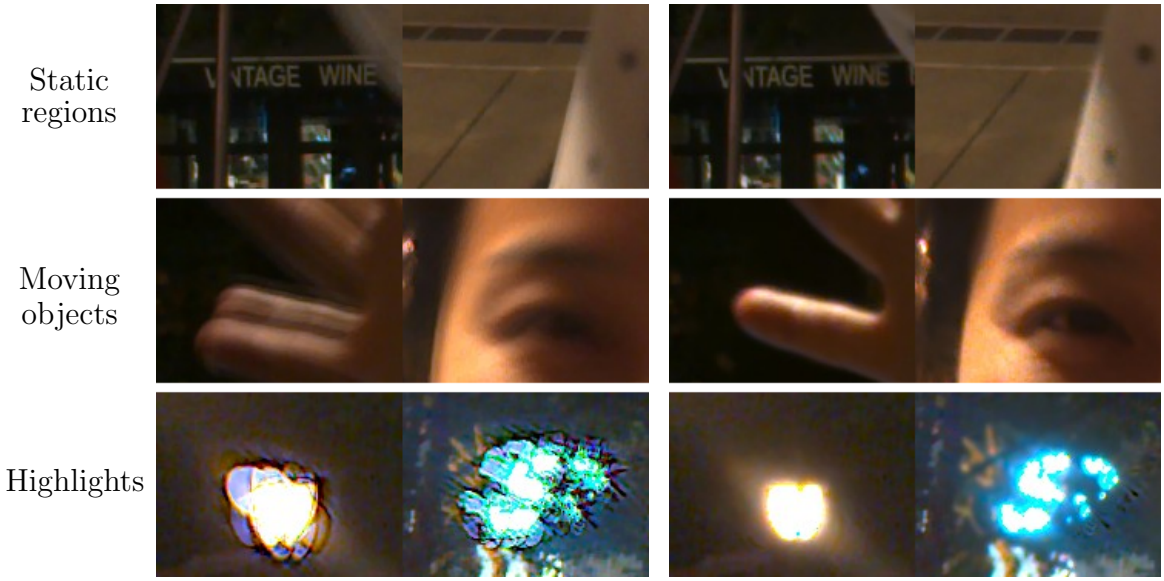


(b) Blend weights for (a)

Figure 4.7: Detecting moving objects and over-exposed regions. A burst of eight images are processed to handle outliers after image deblurring. Two result images are shown in (a) and their blend weights are presented in (b). Note that outliers are effectively detected in (b). Six insets represented in yellow boxes are shown in Figure 4.8 for further comparison.



(a) Reference input $Y_{i_{ref}}$ (b) Denoised image $\widetilde{X}_{dn}$



(c) Deblurred image $\widetilde{X}_{dc}$ (d) Blended image $\widetilde{X}$

Figure 4.8: Output images after handling moving objects and over-exposed regions. Six insets of examples in Figure 4.7 are shown. When multi-image deconvolution is applied, moving objects may suffer from excessive motion blur while over-exposed regions cause artifacts as shown in (c). To hide these artifacts, a patch-based method is applied to a reference image in (a) to get a denoised image shown in (b). Then, (b) and (c) are blended to obtain a result image in (d) by using the blend weight in Figure 4.7b. Note that the results effectively hide artifacts while preserving sharpness in the background.

Chapter 5

Performance of Multi-Image Deconvolution

In the previous chapters, we discussed the implementation details of our gyro-based multi-image deblurring system and showed how our system can recover a clean and sharp image from a set of blurry images and gyroscope measurements. We now move on to the second part of this dissertation, which aims to answer following questions: How well will our deblurring be done with given capture parameters? How do we compare different capture strategies and choose the one that will give best results?

The first step of answering these questions requires defining how the performance of multi-image deconvolution can be modeled with given camera shake. In addition, a method is required to estimate what will be the performance when a certain set of capture parameters is used. Then, the performance can be compared between different capture strategies. However, estimating the performance of image deblurring is a very challenging task because two factors are mixed together: camera shake blur and the scene. Although many image prior models are proposed, it is still hard to model the characteristics of the scene and estimate their influence on deconvolution.

In this chapter, we propose a way of modeling the performance of multi-image deconvolution by using a noise amplification factor. The noise amplification factor quantifies the amount of noise that is amplified by deconvolution. By understanding the properties of this factor, we can describe the performance in terms of exposure

time. Also, we prove an important property of noise amplification factor: *Average noise amplification factor due to camera shake blur increases linearly with exposure time*. Chapter 6 will explore the ramifications of this fact.

5.1 Non-Blind Multi-Image Deconvolution Based on Wiener Estimation

When the blur kernel is available, the latent image can be recovered via non-blind multi-image deconvolution. Wiener deconvolution [61] is a standard frequency domain method for estimating a sharp image from multiple blurry observations. Although there exists other methods that are known to produce visually appealing results, the Wiener deconvolution allows us to provide a theoretical analysis to model its performance.

Wiener deconvolution estimates a latent image X from n blurry images $\{Y_i\}_{i=1,\cdots,n}$ where each blurry image is formulated as in Equation 3.11. By minimizing the expectation of squared estimation error, $\mathrm{E}\left(|\bar{\mathcal{X}} - \mathcal{X}|^2\right)$, in the frequency domain, we get

$$\bar{\mathcal{X}} = \frac{\sum_{i=1}^{n} \frac{1}{\sigma_{\mathcal{N}_i}^2} \mathcal{K}_i^* \mathcal{Y}_i}{\sum_{k=1}^{n} \frac{1}{\sigma_{\mathcal{N}_k}^2} |\mathcal{K}_k|^2 + \frac{1}{|\mathcal{X}|^2}} \tag{5.1}$$

where $\bar{X}$ is the optimum estimate of X, K_i is the known ith blur kernel, n is the number of images, $\mathcal{X}$, $\bar{\mathcal{X}}$, $\mathcal{K}_i$, and $\mathcal{Y}_i$ denote the Fourier transforms of X, $\bar{X}$, K_i, and Y_i, respectively, $|\mathcal{X}|^2$ is the power spectrum of X, $\mathcal{N}_i$ is the white Gaussian noise in the frequency domain with variance $\sigma_{\mathcal{N}_i}^2$. Note that $\sigma_{\mathcal{N}_i}^2 = n_u n_v \sigma_{N_i}^2$, where $\sigma_{N_i}^2$ is the noise variance of Y_i in the spatial domain, n_u and n_v are the numbers of pixels in each dimension. This estimation can be interpreted as the weighted average of individual estimates that are obtained from each image, and their weights are determined by input noise variance and the attenuation in the frequency domain caused by each blur. The image statistics, $|\mathcal{X}|^2$, is also considered in the estimation to suppress undesirable noise amplification.

By using the Parseval's theorem, total MSE between X and $\bar{X}$ over the whole image can be calculated either in the spatial domain (x, y) or in the frequency domain (u, v):

$$\mathrm{MSE}(\bar{X}) = \frac{1}{n_x n_y} \sum_{x,y} \mathrm{E}\left(|\bar{X} - X|\right)^2 = \frac{1}{n_u^2 n_v^2} \sum_{u,v} \mathrm{E}\left(|\bar{\mathcal{X}} - \mathcal{X}|\right)^2 \tag{5.2}$$

where n_x and n_y are dimensions in the spatial domain and are equal to n_u and n_v respectively. This estimation error is derived in the frequency domain as

$$\mathrm{MSE}(\bar{X}) = \frac{1}{n_u^2 n_v^2} \sum_{u,v} \frac{1}{\sum_{k=1}^{n} \frac{1}{\sigma^2_{\mathcal{N}_k}} |\mathcal{K}_k|^2 + \frac{1}{|\mathcal{X}|^2}}. \tag{5.3}$$

Note that the denominator consists of two terms: power spectra of blur kernels and the image statistics.

5.2 Derivation of Noise Amplification Factor

To get some insight into the deconvolution procedure in Section 5.1, let us make two assumptions about the deconvolution and derive a noise amplification factor. First, let us assume all input images are captured with the same exposure time t and analog gain g. In this case, the noise statistics for all images are identical, $\{\sigma^2_{N_i}\}_{i=1,\cdots,n} = \sigma^2_N$. Second, no prior on the image statistics is assumed. This is the case for an unbiased optimal estimator. The estimation no longer depends on the scene, and the term $\frac{1}{|\mathcal{X}|^2}$ in Equation 5.1 and Equation 5.3 can be ignored.

When these two assumptions hold, Equation 5.3 reduces to

$$\mathrm{MSE}(\bar{X}) = \gamma \frac{\sigma^2_N}{n} \tag{5.4}$$

where γ is the noise amplification factor defined as

$$\gamma = \frac{1}{n_u n_v} \sum_{u,v} \frac{1}{\frac{1}{n} \sum_{i=1}^{n} |\mathcal{K}_i(u,v)|^2}. \tag{5.5}$$

The main idea of the formulation of Equation 5.4 is that the estimation error is decomposed into two separate terms: camera shake blur and input noise. The first term, the noise amplification factor, depends only on the characteristics of the blur applied to the system. The second term represents input noise variance that is averaged over n images, and this term denotes the amount of noise when blur is negligible. Note that γ quantifies the increase in high frequency noise when average power spectrum of the blur kernels is inverted in the deconvolution.

Figure 5.1 shows an example of multi-image deconvolution when a burst is captured with the same exposure parameters and no image prior is assumed for the deconvolution. The power spectra of eight blur kernels shown in Figure 5.1a are averaged to obtain the average power spectrum in Figure 5.1d. We estimate γ by averaging Figure 5.1e, which is the inversion of the average power spectrum, over the whole frequency domain. This example demonstrates each step of evaluating Equation 5.5.

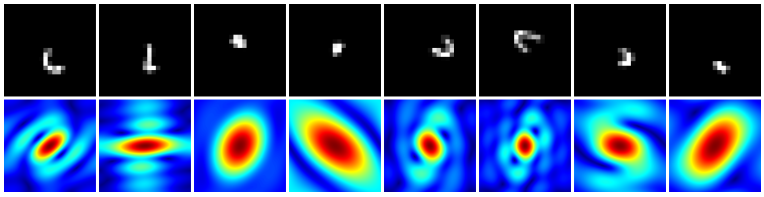
When the optimal estimate in Equation 5.1 is used, the SNR of the estimate can be obtained as

$$\mathrm{SNR}(\bar{X}) = \frac{\sqrt{n}}{\sqrt{\gamma}} \frac{X}{\sigma_N}. \tag{5.6}$$

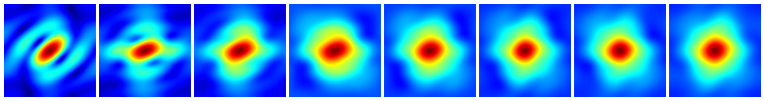
By substituting our image noise model described in Equation 3.4, we have

$$\mathrm{SNR}(\bar{X}) = \frac{\sqrt{n}}{\sqrt{\gamma}} \frac{c_1 g t \Phi}{\sqrt{c_2 g^2 t \Phi + g^2 \sigma_{r_0}^2 + \sigma_{r_1}^2}}. \tag{5.7}$$

We use this SNR as a measure that quantifies the performance of our image deconvolution method in the following analysis. If we can relate γ with exposure parameters, the performance can be estimated as a function of exposure parameters. Thus, understanding the properties of γ is an essential task, which will be presented in the next section.

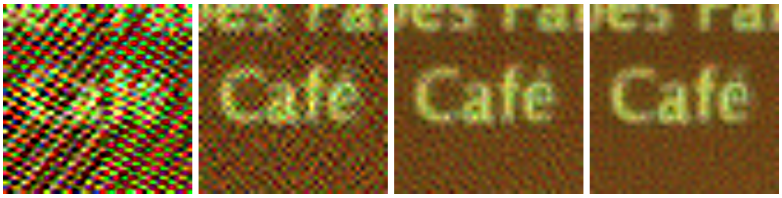


(a) A set of blur kernels and their power spectra



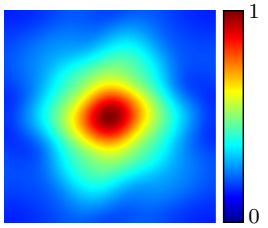
n=1 n=2 n=3 n=4 n=5 n=6 n=7 n=8

(b) Average power spectrum of n images

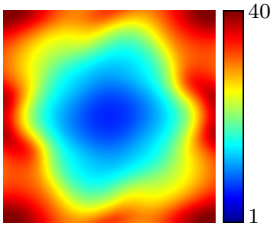


n=1 n=2 n=4 n=8

(c) Multi-image deconvolution applied to n images



(d) $\frac{1}{n}\sum_{i=1}^{n}|\mathcal{K}_i|^2$



(e) $1/\left(\frac{1}{n}\sum_{i=1}^{n}|\mathcal{K}_i|^2\right)$

Figure 5.1: An example of n-image deconvolution. Eight images are captured from a single instrumented hand-held capture. Subfigure (a) shows eight consecutive blur kernels, $\{K_i\}_{i=1,\cdots,n}$, and their power spectra, $\{|\mathcal{K}_i|^2\}_{i=1,\cdots,n}$, respectively. Multi-image deconvolution is applied using n images without any prior, to show the improvement when more images are used. As the power spectra in (b) show, adding more images fills in missing frequency components. Although no prior is used, the output images in (c) are not severely degraded by noise when more than two images are used. In addition, each step of evaluating the noise amplification factor γ for this example is demonstrated when $n = 8$. The eight power spectra in (a) are averaged as shown in (d), and the plot shown in (e) is obtained by taking the inverse of the average power spectrum. Then, this plot is averaged over the whole frequency domain to get $\gamma = 15.96$. Note that the evaluation of this factor represents the deconvolution process, and depends only on the given blur kernels.

5.3 Modeling the Noise Amplification Factor with Exposure Parameters

In this section we focus on understanding handshake blur in real-world situations. Our main tool for this analysis will be the noise amplification factor defined in Section 5.2, and we show how this factor is related to exposure parameters. The main challenge is that the blur is affected by various factors: exposure settings (exposure time, frame-to-frame time, and number of shots), user, type of camera (weight, grip type, optics, and pixel size), and so on. In addition, camera shake blur is always different from capture to capture, and thus, a large number of samples is required to find meaningful statistics of the blur. To explore this high-dimensional parameter space, we use simulation to build an empirical model based on real measurements of camera shake.

As a start, we built a database of camera shake by collecting 100 sequences of real camera motion that are measured by a gyroscope. Each user was asked to hold a tablet steady for a 20-second exposure, while images and gyroscope streams are recorded. This database allows us to create independent samples of camera motion by randomly selecting a small slice from a gyroscope stream. Figure 5.2 demonstrates the case when $n = 8$ images are captured for given t_{exp} and t_{ft}, and the start time of first exposure t_s^1 is randomly selected. Then, we can simulate various image capture scenarios, and evaluate $\bar{\gamma}$, the noise amplification factor averaged over different camera shakes.

Figure 5.3 shows how $\bar{\gamma}$ is related to exposure time under various combinations of parameter settings. As shown in Figure 5.3a, $\bar{\gamma}$ is close to 1 when t_{exp} is very small because no blur exists. For longer exposures, $\bar{\gamma}$ increases with exposure time as image becomes more blurry and more noise is amplified by deconvolution. If more number of images is captured, $\bar{\gamma}$ is reduced because missing frequency components in an image can be preserved in other images. Figure 5.1a shows that different blur preserves different frequency information of the scene. The blue regions in the power spectra visualize missing frequency components that may cause severe artifacts after deconvolution. When multiple images are available, average power spectrum of these

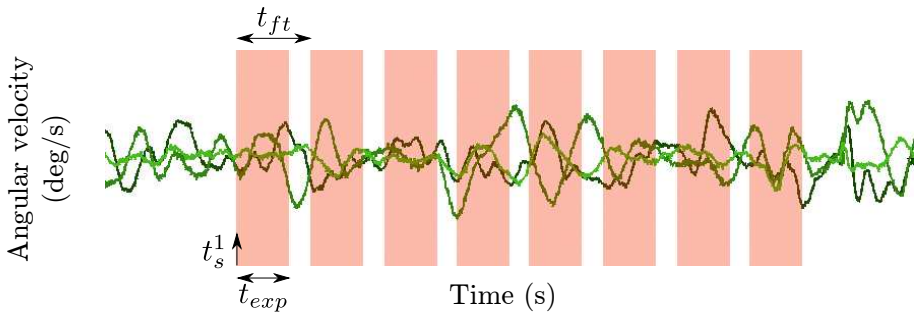


Figure 5.2: Camera motion simulation for different exposure parameters. From a gyroscope measurement for a relatively long period, numerous independent camera motion samples can be simulated for different capture strategies. For given n, exposure time t_{exp} and frame time t_{ft}, the start time of first exposure t_s^1 can be randomly selected to generate a camera motion sample. The time windows in red represent exposure windows when $n = 8$ images are captured.

blur kernels contains less amount of blur region as shown in Figure 5.1b, and the deconvolution becomes more stable.

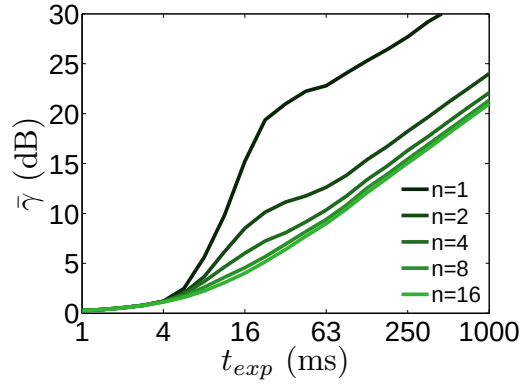
The important observation is that $\bar{\gamma}$ grows *linearly* with t_{exp} as shown in Figure 5.3b: the ratio between $\bar{\gamma}$ and t_{exp} remains constant when the exposure time per shot is long enough. This relation is more obvious when n is large, and is consistently observed among different users (Figure 5.3c), or when different frame times (Figure 5.3d) or focal lengths of camera optics (Figure 5.3e) are used.

We can explain this trend mathematically for large n and t as follows. Let us define the average power spectrum of blur kernels as

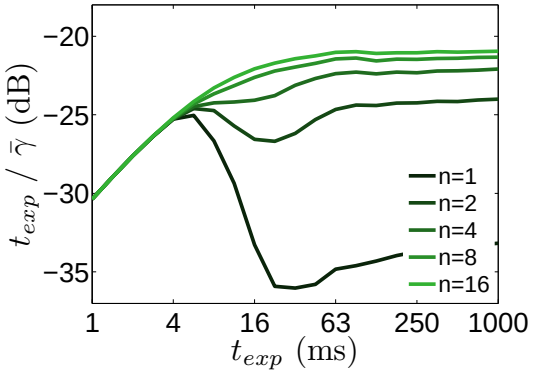
$$\bar{P}_t = \frac{1}{n_i} \sum_i |\mathcal{K}_{i,t}|^2 \tag{5.8}$$

where $K_{i,t}$ denotes ith sample of blur kernel for exposure time t, $\mathcal{K}_{i,t}$ is the Fourier transform of $K_{i,t}$, n_i is the number of samples, and assume $\bar{P}_t$ converges for large n_i. Let us think of an image as the sum of first- and second-half images with the same exposure time of $\frac{1}{2}t$. Then, we can split the blur kernel $K_{i,t}$ into two as

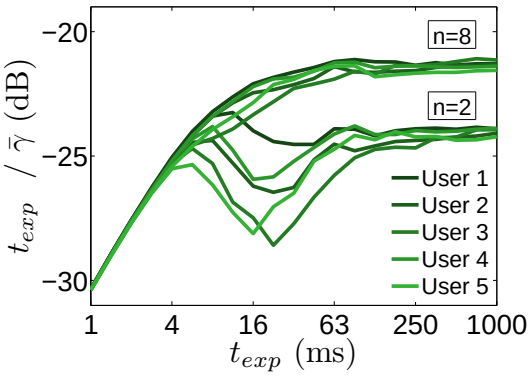
$$K_{i,t} = \frac{1}{2}\left(K_{i_1,\frac{t}{2}} + K_{i_2,\frac{t}{2}}\right) \tag{5.9}$$



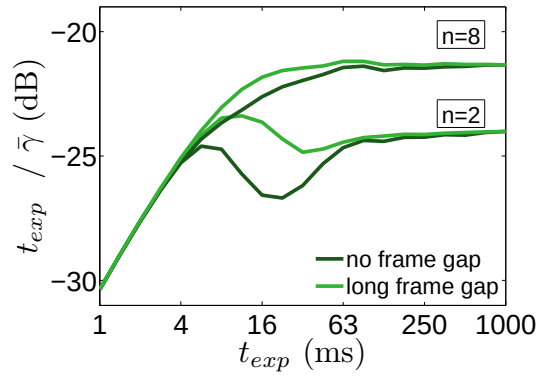
(a) Number of shots n



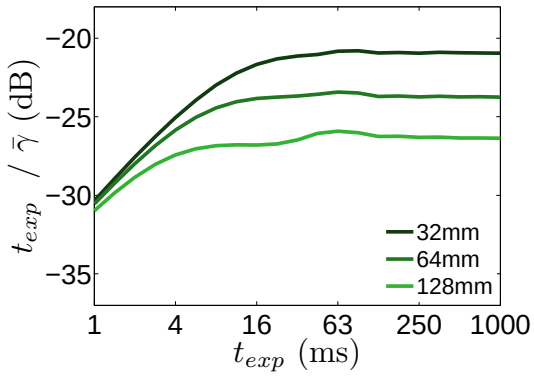
(b) Number of shots n



(c) User

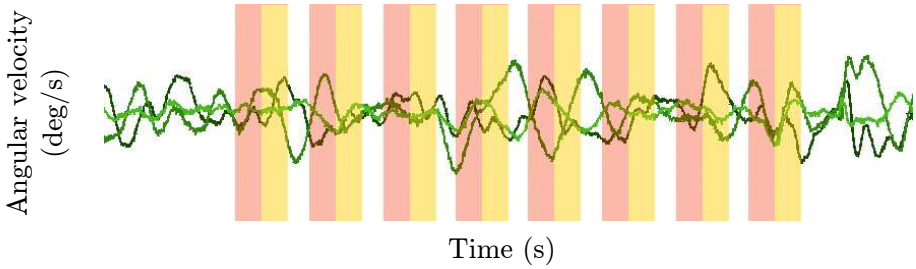


(d) Frame time t_{ft}

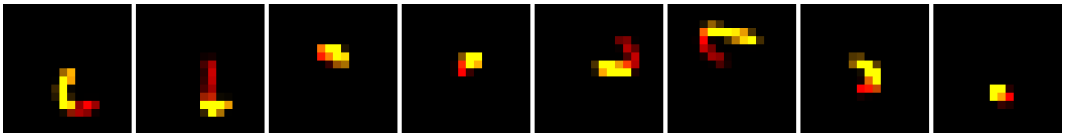


(e) Focal length f

Figure 5.3: Average noise amplification factor measured from simulation based on real camera motion. Each plot shows how $\bar{\gamma}$ varies when a different parameter is changed as denoted by the plot captions. Subfigure (a) shows $\bar{\gamma}$ while (b)-(e) plot the ratio between $\bar{\gamma}$ and per-image exposure time t_{exp}. Note that the ratio becomes constant for long exposures, which means that $\bar{\gamma}$ increases linearly with t_{exp}.



(a) Splitting exposures into first- and second-half exposures



(b) Splitting blur kernels

Figure 5.4: Splitting exposures into first- and second-half exposures. (a) An example of exposure windows are shown when $n = 8$. Each exposure is equally split into two exposures. The exposures in red represent first-half exposures and the yellow regions show second-half exposures. (b) Each corresponding blur kernel $\{K_{i,t}\}_{i=1,\cdots,n}$ is also split into red and yellow kernels that represent $\{K_{i_1,\frac{t}{2}}\}_{i=1,\cdots,n}$ and $\{K_{i_2,\frac{t}{2}}\}_{i=1,\cdots,n}$, respectively.

where i_1 and i_2 denote the indices corresponding to first- and second-half of i, respectively. Then, the power spectrum of $K_{i,t}$ can be approximated as

$$|\mathcal{K}_{i,t}|^2 = \frac{1}{4}\left(\left|\mathcal{K}_{i_1,\frac{t}{2}}\right|^2 + \left|\mathcal{K}_{i_2,\frac{t}{2}}\right|^2 + 2\mathrm{Re}\{\mathcal{K}_{i_1,\frac{t}{2}}\mathcal{K}^*_{i_2,\frac{t}{2}}\}\right) \tag{5.10}$$

$$\approx \frac{1}{4}\left(\left|\mathcal{K}_{i_1,\frac{t}{2}}\right|^2 + \left|\mathcal{K}_{i_2,\frac{t}{2}}\right|^2\right) \tag{5.11}$$

if we assume the crosscorrelation term is negligible because consecutive blur kernels become uncorrelated for relatively long t. Then, we get the recursive relation for average power spectrum $\bar{P}_t = \frac{1}{2}\bar{P}_{\frac{t}{2}}$, which leads to $\bar{\gamma}_{n,t} = 2\bar{\gamma}_{n,\frac{t}{2}}$ for large n. This shows the linear relation for large n and t:

$$\bar{\gamma}_{n,t} \propto t. \tag{5.12}$$

To verify the assumption that blurs are uncorrelated for large t, we evaluate $\bar{\gamma}$

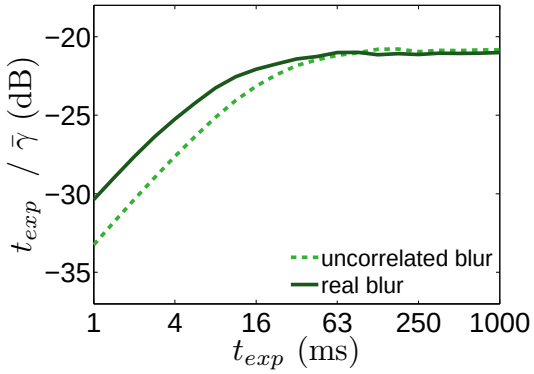


Figure 5.5: Uncorrelated blur for long exposure times. The average noise amplification factor $\bar{\gamma}$ is evaluated by assuming that consecutive blur kernels are uncorrelated to each other. The result obtained from the approximation agrees well with real measurements when t_{exp} is longer than ∼50ms.

by using the approximation for $n = 16$ and compare it with our evaluation shown in Figure 5.3b. When the approximation is adopted, blur kernels are split into two sets. Figure 5.4 shows an example of these two sets of blur kernels that correspond to first- and second-half exposures. Then, the average power spectrum in the denominator of Equation 5.5 is evaluated by following the approximation described in Equation 5.11. Figure 5.5 shows the results. When exposure time is short, the assumption does not hold because the shape of blurs is linear and consecutive blur kernels are strongly correlated. However, when exposure time is longer than ∼50ms, blurs become uncorrelated and the result obtained from the approximation agrees well with real measurements. This reduced correlation at time scales longer than 1/20 second makes sense, because human muscular handshake has a period of about 10 fps.

Figure 5.3d shows another effect that uncorrelated blurs have on $\bar{\gamma}$: uncorrelated blurs reduce $\bar{\gamma}$ compared to correlated ones. Uncorrelated blurs preserve frequency components that are complementary to each other while correlated ones only repeat similar observations. Longer frame-to-frame time makes the blurs less correlated, which results in smaller $\bar{\gamma}$ as shown in Figure 5.3d.

In summary, when multi-image deconvolution is applied, we observed that the amount of noise amplification increases linearly with exposure time for large n and t. This observation comes from the fact that camera shake blurs become uncorrelated

to each other when exposure time is long enough. Note that we can generally assume blurs are uncorrelated after a certain time for any camera or any user. Thus, the linear relation in Equation 5.12 holds for *any* camera shake for large n and t.

Chapter 6

Analysis of Alternative Capture Strategies

Chapter 5 discussed how the performance of multi-image deconvolution can be quantified by using a noise amplification factor. The characteristics of this noise amplification factor are analyzed to find its relation to various capture parameters. From the analysis, we proved that this factor increases linearly with exposure time for any camera shake.

In this chapter, we put previous analyses together and finally present how well multi-image deconvolution will work in various photographic situations. We show an important property of multi-image deconvolution: Multi-image deconvolution does not benefit from increasing exposure time beyond a certain threshold. This property plays a key role in determining good data sets to capture in low-light situations. Then, the performance of multi-image deconvolution is compared to align-and-average by performing simulations on real scenes. Finally, based on the comparison, guidelines are presented to help users to determine good capture strategies in real-world situations.

no blur	large blur
$\bar{\gamma} = 1$	$\bar{\gamma} \propto t$

(a) Noise amplification factor

read noise	photon noise
$\sigma_N^2 = c$	$\sigma_N^2 \propto t$

(b) Noise characteristics

	no blur ($\bar{\gamma} = 1$)	large blur ($\bar{\gamma} \propto t$)
read noise dominant	SNR $\propto \sqrt{n}gt$	SNR $\propto \sqrt{n}g\sqrt{t}$
photon noise dominant	SNR $\propto \sqrt{nt}$	SNR $\propto \sqrt{n}$

(c) SNR for different photographic situations

Table 6.1: Summary of the performance of multi-image deconvolution in various capture situations. We assume that read noise before amplification is negligible.

6.1 Performance of Capture Strategies

When multi-image deconvolution is applied to a burst of images captured from a hand-held device, we showed that the SNR of the deconvolved result can be expressed as in Equation 5.7. As assumed in this analysis, we first consider the case when no image prior is used in the deconvolution. Then, the analysis is extended to the case where image priors are used.

To begin, let us consider different photographic situations to get more insight into the relation between the SNR and capture parameters. Table 6.1 shows four different situations depending on the property of the noise amplification factor and input noise variance. As shown in Section 5.3 and Table 6.1a, the noise amplification factor increases linearly with exposure time for long exposures, while it is close to 1 when blur is negligible in captured images. The relation between input noise variance and exposure time is determined by the dominant source of noise as described in Section 3.1.1. As shown in Table 6.1b, when photon noise is dominant, the variance increases with exposure time. In contrast, the amount of noise is fixed when read noise dominates. Here, we assume that read noise before amplification is negligible.

The relationships between output SNR and various capture parameters can be obtained for each situation, and are summarized in Table 6.1c. For all cases, the SNR increases with the square root of the number of images captured. Taking additional

images is always beneficial, but in turn, total capture time is increased. On the contrary, the effect of changing exposure time depends on the photographic situations. Note that if no blur exists (first column of Table 6.1c), then the analysis reduces to the case of align-and-average. When photon noise is dominant and large blur is involved (lower right cell of Table 6.1c), then the SNR does not improve with exposure time. This is an unexpected result because it is commonly thought that increasing exposure time helps to get images with better quality. This common thought is valid only when there is no handshake blur. When a hand-held device is used, the noise introduced by computationally removing additional blur cancels the improvement in SNR due to reduced shot noise in input images. In other words, longer exposures are preferable, at least to the point where handshake blur becomes intolerable, and if read noise is dominant, then this transition occurs at a longer exposure time. This property is already observed in Figure 5.3. The y axis, the ratio between $\bar{\gamma}$ and exposure time, represents the SNR and the curves in the plot become flat when exposure time is long enough.

6.2 The Effect of Image Priors

The analysis in the previous section is only based on the camera shake blur applied to the system, without considering the scene contents. Although we considered the scene brightness to model the properties of image noise, real-world scenes are more complicated to model and the results are strongly affected by the choice of image prior model used in deconvolution.

In this section, we empirically show what effects image priors might have in the analysis. We compare two approaches in removing camera shake: align-and-average (AA) and multi-image deconvolution (MD). Both cases are tested with and without assuming image priors. To make a fair comparison between AA and MD when image priors are used, an additional denoising step is applied on the output of AA by applying Equation 4.2 with $\mathcal{K}_i = 1$ and the Gaussian prior to suppress the same amount of noise as in MD. Given capture parameters, a burst of blurry images is simulated using real measurements of camera motion as described in Section 5.3.



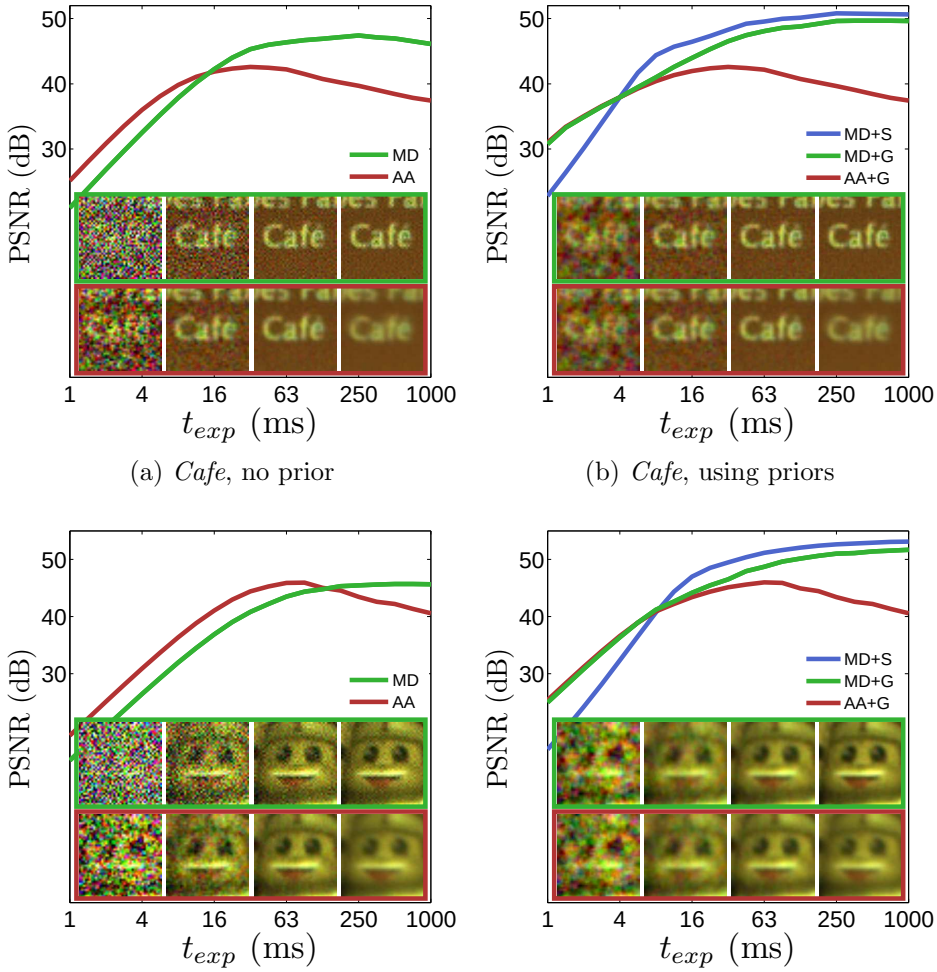
(a) *Cafe* (b) *Office*

Figure 6.1: Two scenes used for simulations with image priors. Insets shown in yellow boxes are presented in Figure 6.2.

These blurry images are jointly deconvolved, assuming that exact camera shake blur is known. The performance is evaluated by measuring the peak signal-to-noise ratio (PSNR) with respect to the ground truth scene, and the PSNR is averaged over many trials with different camera motions.

Two examples shown in Figure 6.1 are used for simulation when eight images are captured with a unity analog gain, and Figure 6.2 shows the simulation results. The PSNR of AA drops for long exposures because of the handshake blur, while the performance of MD remains stable. Using image priors usually gives better PSNR than the case without image priors, and the improvement in PSNR of MD approaches a horizontal asymptote for both cases, with and without image priors, as exposure time becomes sufficiently long. Introducing a prior improves the PSNR, and it does not significantly change the relation to exposure time as shown in Table 6.1c. As a result, an analysis based on the noise amplification factor can still help estimate the performance of MD even when image priors are used.

Also, note that when the same parameters and image prior are used, MD performs better than AA as shown in Figure 6.2b and Figure 6.2d for the Gaussian prior. This is not surprising as MD is a more sophisticated approach to handle blurry images and is expected to work better when the same input data is used. However, if different



(a) *Cafe*, no prior

(b) *Cafe*, using priors

(c) *Office*, no prior

(d) *Office*, using priors

Figure 6.2: The performance of align-and-average (AA) and multi-image deconvolution (MD) with image priors. Two example scenes in Figure 6.1 are used for simulation. The curves are similar to Figure 5.3, but this time with priors. Subfigures (a)(c) are obtained with no prior used, while (b)(d) show the improvement when image priors are employed. The sparse prior (S) shown with the blue curve gives slightly more improvement than the Gaussian prior (G) in green. A small region of output images for different exposure times is shown. The insets in red boxes correspond to the output of AA while green boxes correspond to the output from MD. The performance follows the analysis in Table 6.1c even when image priors are used.

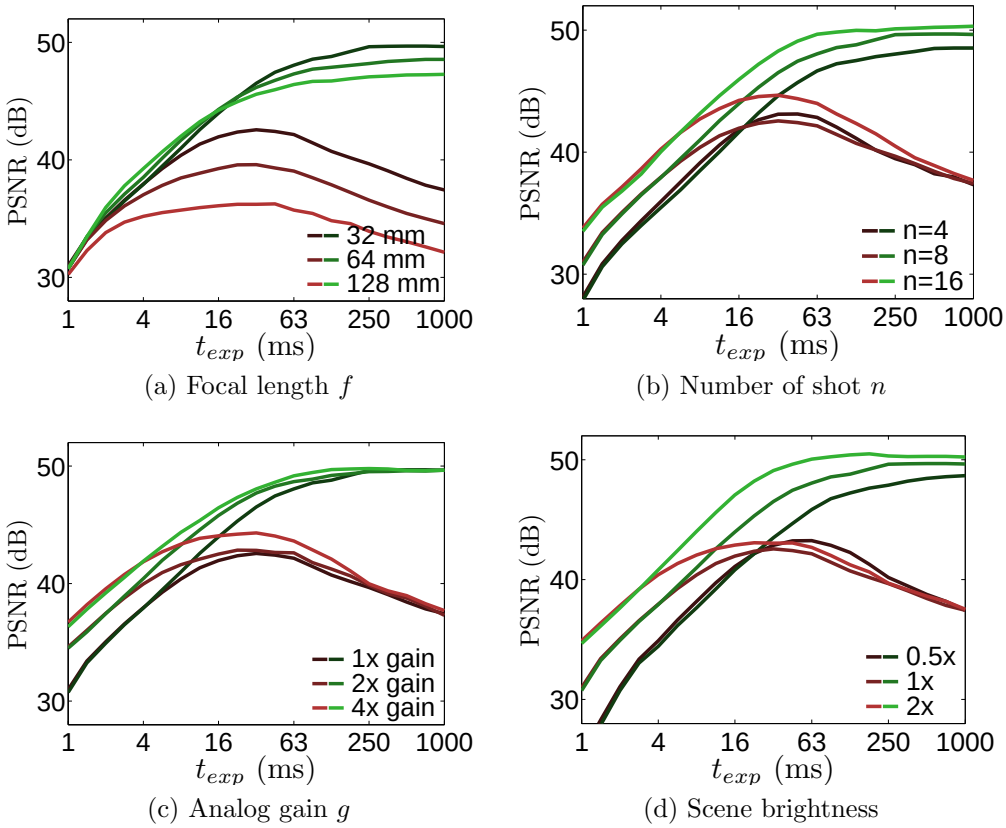
parameters are used to capture the images for AA and MD, comparing the performance of two approaches becomes more complicated. Some examples of this case will be discussed in Section 6.4.

6.3 The Effect of Various Parameters

From the previous sections, we showed that two factors mainly affect the performance of multi-image deconvolution: the amount of camera shake blur and the characteristics of image noise in captured images. These factors are dependent on given photographic situation, which is described by various parameters. We assume the user has control of n, g, and t_{exp}, and faces variability in scene statistics and dynamic range, and camera noise model, optics, and so on. To understand the effect of each factor, we performed simulation on the *Cafe* scene in Figure 6.1a, changing one axis of the parameter space at a time. The simulation results are shown in Figure 6.3.

Let us consider the case when focal length of the optics is changed. Focal length is one of the main factors that determine the amount of camera shake blur. When longer focal lengths are used, camera motion is amplified more by the narrow field of view. As shown in Figure 6.3a, the performance of AA is significantly degraded with long focal lengths because the remaining blur after AA is no longer tolerable.

Other parameters mainly affect the properties of input image noise. Capturing more images reduces both photon noise and read noise, while making deconvolution stable at the same time (Figure 6.3b). Using higher analog gain helps to suppress read noise but does not affect the amount of photon noise (Figure 6.3c). Brighter scenes irradiate more photons and thus give more photon noise (Figure 6.3d). The combined effect of all these parameters determine the exposure time when photon noise is dominant. This exposure time is important to estimate when MD will not improve with exposure time anymore.



(a) Focal length f

(b) Number of shot n

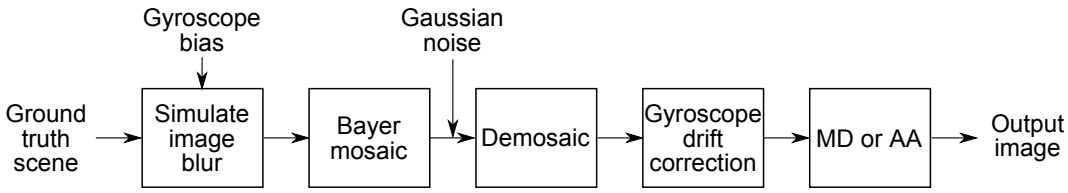
(c) Analog gain g

(d) Scene brightness

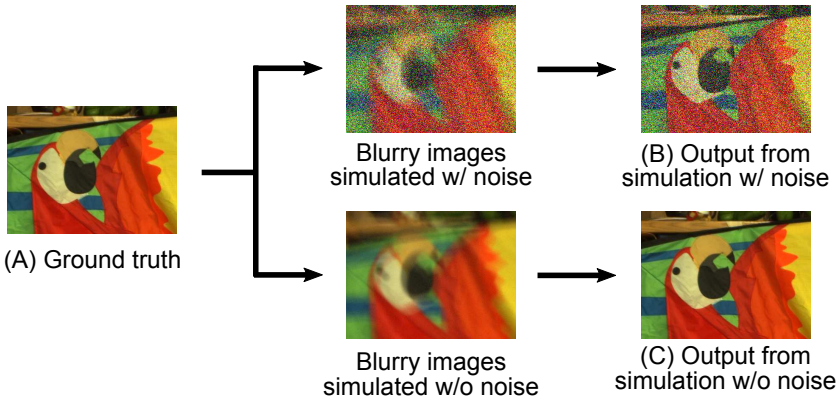
Figure 6.3: The effect of changing each parameter on the performance of align-and-average (red) and multi-image deconvolution (green). The results are from simulation, and all cases assume the Gaussian prior. Focal length strongly affects the range where align-and-average is effective as shown in (a). Other parameters shown in (b)-(d) change the ratio between read noise and photon noise.

6.4 Comparing Capture Strategies in Real-World Situations

In this section, our analysis is extended to handle various practical issues that exist in real-world scenarios. This is done by performing extensive simulations that simulate the whole processing pipeline from image formation to gyroscope drift correction and multi-image deconvolution as shown in Figure 6.4a. This simulation pipeline aims to produce more realistic image noise that contaminates captured images. Rather than



(a) Our simulation pipeline



(b) Using two simulation results

Figure 6.4: A simulation pipeline for comparing capture strategies. (a) Our simulation is designed to incorporate various practical issues that are common in real-world situations. (b) The simulation is repeated twice, with and without applying noise. Then, the amount of noise and remaining blur can be quantified separately. The noise is represented by the PSNR between B and C, while the sharpness is estimated using the SSIM [59] between A and C.

adding independent Gaussian noise in the RGB space, the Gaussian noise is added to the Bayer images followed by demosaic operation. Image noise becomes strongly correlated between color channels and usually shown as color blobs or structured patterns after demosaicing. This correlated noise is more difficult to handle [45]. In addition, gyroscope drift correction is performed in the simulation pipeline. This correction can be considered equivalent to finding an alignment between input images, and its accuracy is critical to obtain compelling results from MD and AA. This simulation pipeline is repeated with different camera motions to obtain a meaningful statistics of each capture strategy for given capture parameters.

The main challenge in comparing different strategies is that there exists a trade-off between noise and blur. Thus, it is hard to decide whether we want to prefer

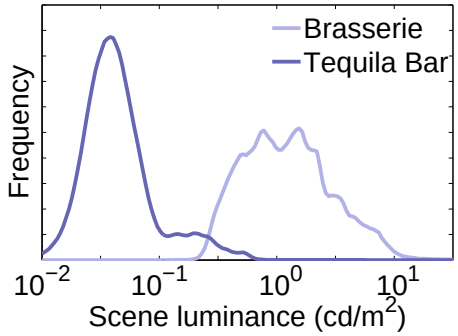
noisy sharp images or slightly blurry ones with less noise. In addition, measuring the amount of noise or blur is not trivial because both factors affect image quality metrics at the same time. The SNR, for example, only measures the combined effect of noise and blur, with no control on which factor to consider more important. Also, most image sharpness metrics cannot distinguish image structures from noise and often prefer images with higher noise levels.

To solve this problem, we propose to use two simulation results as shown in Figure 6.4b. Our simulation pipeline is repeated twice, with and without applying noise when input images are formulated. When the noise is not applied in the simulation, output images will be clean while containing the same amount of blur that remains after MD or AA. Then, the amount of noise can be estimated separately from the remaining blur by evaluating the PSNR between the output images obtained from two simulations. In addition, the amount of remaining blur is quantified by measuring a full-reference image quality metric, structural similarity (SSIM) [59], between the ground truth and the output image obtained from the simulation without noise. Because the output image contains little noise, the metric can be robustly measured. We use these two metrics, PSNR and SSIM, to describe the trade-off space that represents blur and noise.

6.4.1 Simulation Results: An n-Shot Burst

Let us begin with the case when the same number of images is captured for different strategies. We mainly focus on comparing the performance of align-and-average (AA) and multi-image deconvolution with the Gaussian prior (MD+G). The relation between the performance and per-image exposure time t_{exp} is discussed in the trade-off space.

Figure 6.6 and Figure 6.7 show the results obtained from the two simulations described in Figure 6.4 when the *Brasserie* scene (Figure 6.5b) is used as the ground truth and an eight-shot burst is captured for each camera motion. A representative case is selected from simulation results that are repeated with different camera motion, and shown in Figure 6.6. When AA is used, the PSNR increases with exposure time,



(a) Scene histogram



(b) *Brasserie*



(c) *Tequila Bar*

Figure 6.5: Two examples of ground truth scenes used for our simulations. The scene histograms in (a) show that the *Tequila Bar* is darker than the *Brasserie*.

as shown in Figure 6.7a. The amount of improvement is more than the case of MD+G because MD tends to amplify noise after deconvolution. However, long exposures are not necessarily desirable for AA because they decrease the SSIM, as shown in Figure 6.7b. In contrast, the SSIM of MD+G is maintained higher than the AA's for long exposures. However, it is also not desirable to use an exposure time longer than a certain point because high frequency components of the scene may have been lost and the gyroscope bias estimation becomes less accurate when handshake blur is intolerable. This relation is visualized in Figure 6.7c. Note that each point on the curve represents the expected performance of a capture strategy, and when different exposure times are considered, the point moves along the trade-off curve. The upper

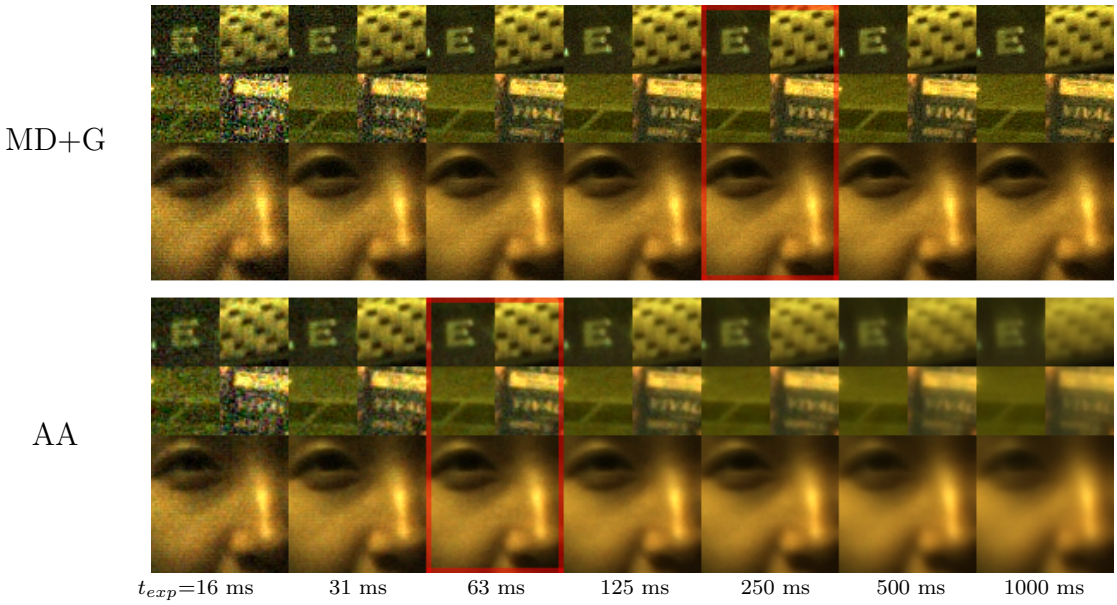
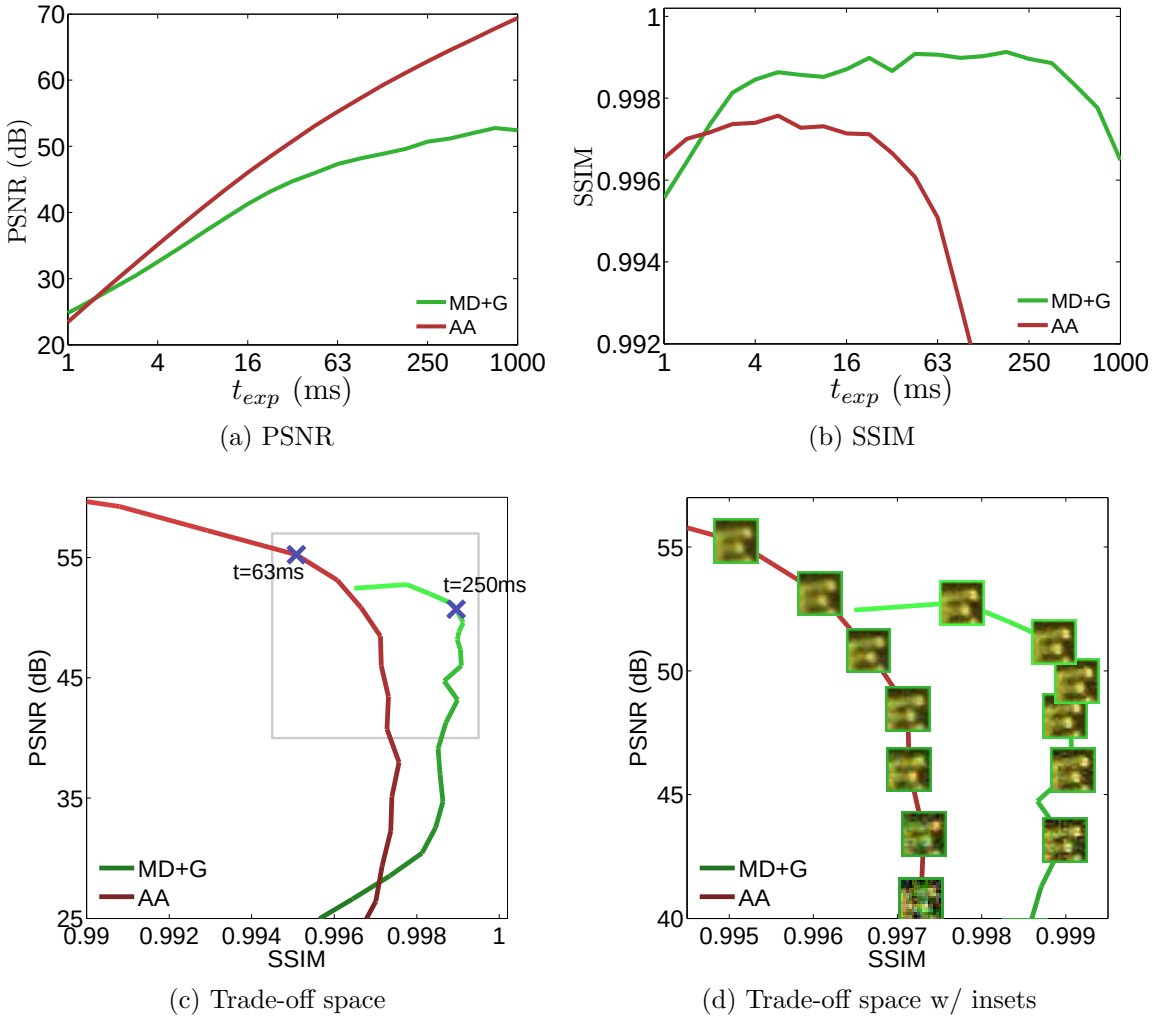


Figure 6.6: Comparison between capture strategies: an n-shot burst (image insets). The *Brasserie* scene is used to simulate captures of an eight-shot burst for different exposure times. Two approaches, AA and MD+G, are applied. Here, five small regions of the output images simulated with a representative camera motion is shown. Note that AA suffers from blur for long exposures. A possible choice of good exposure time for each approach is denoted in a red box. Figure 6.7 shows the plots for these simulation results.

right region of the trade-off space is desirable because it presents capture strategies that give clean and sharp images. Thus, a good exposure time for each approach can be selected from this trade-off curve, and a possible choice is represented in Figure 6.7c. Note that when the same number of images is captured, MD+G usually outperforms AA as discussed in Section 6.2.

One may wonder why the amount of blur contained in Figure 6.6a is less than what users may expect in real-world situations. One reason is that our camera motion is measured when the tablet was held steady, without considering the effect of pressing the shutter button. To get more realistic amount of blur, our experiment is repeated with a focal length that is doubled. Figure 6.8 shows that the SSIM is decreased for both approaches because of the amplified blur. Also, output images for AA in Figure 6.8a contain more blur. Again, from the trade-off curves, we selected two exposure

times that give good results for both approaches, as shown in Figure 6.8a and Figure 6.8b. Note that we reduced the exposure time for AA by a factor of two to avoid introducing more blur.



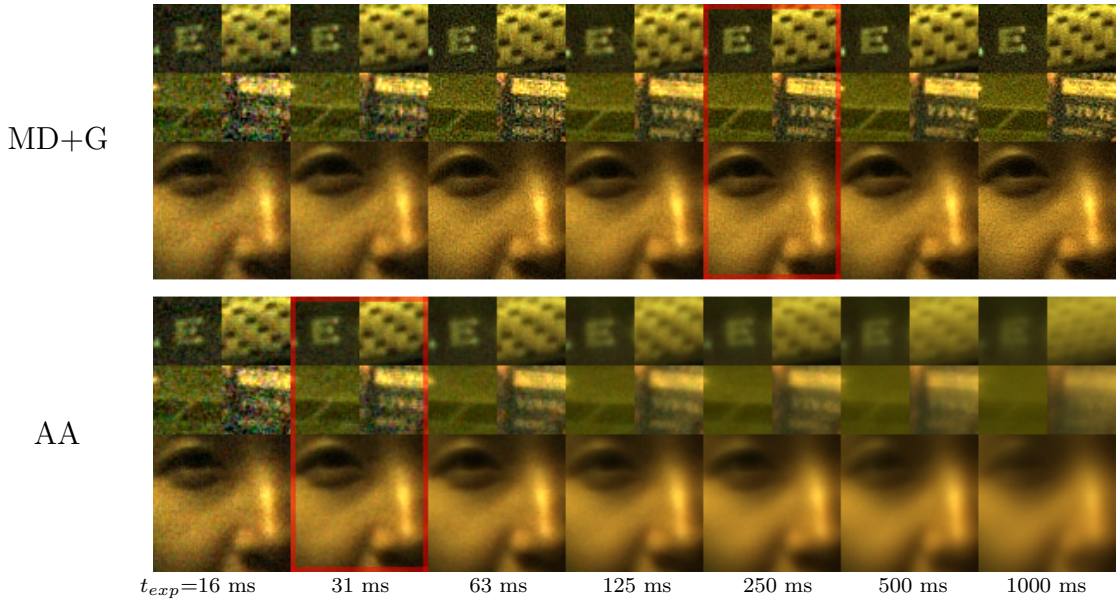
(a) PSNR

(b) SSIM

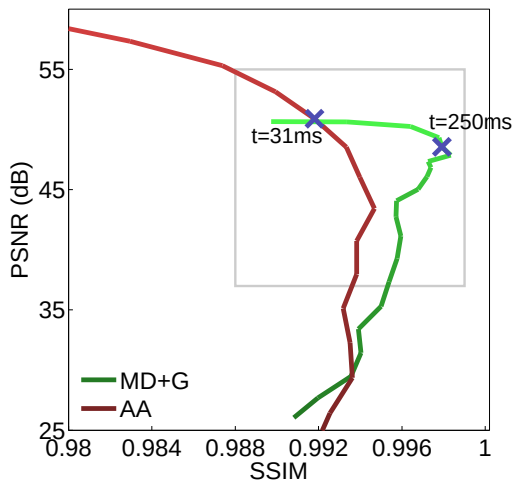
(c) Trade-off space

(d) Trade-off space w/ insets

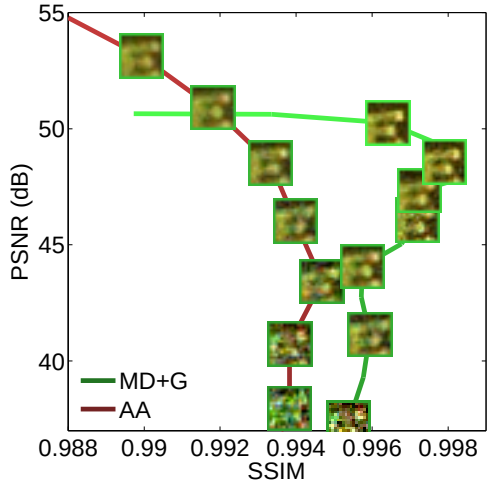
Figure 6.7: Comparison between capture strategies: an n-shot burst (plots). The plots obtained from the experiments in Figure 6.6 are shown. The trade-off curves shown in (c) are obtained by combining plots in (a) and (b). Also, a possible choice of good exposure time for each approach is represented in (c). In (d), a rectangular region in (c) is shown with image insets to visualize the trade-off space.



(a) Image insets



(b) Trade-off space



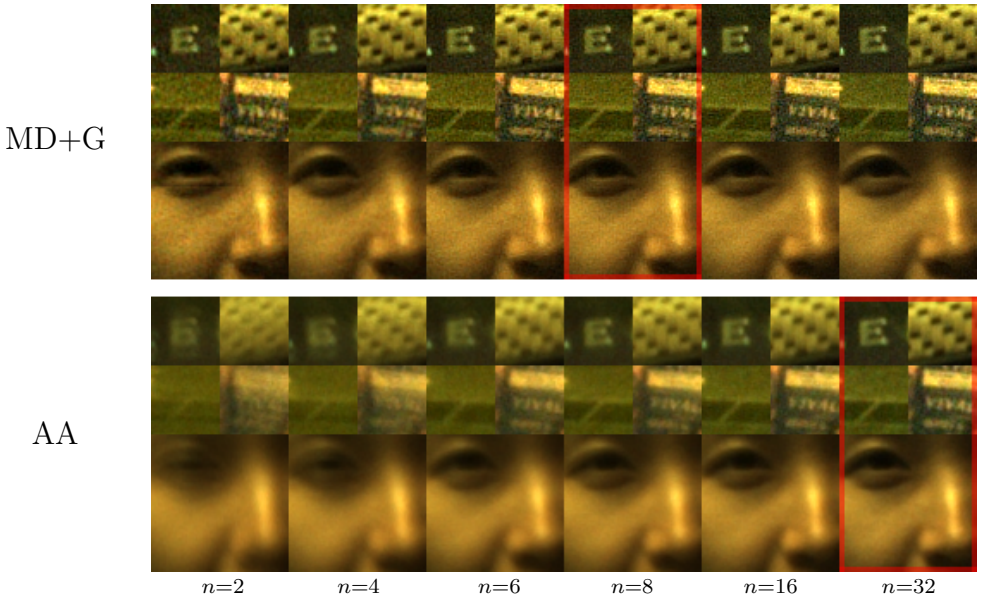
(c) Trade-off space w/ insets

Figure 6.8: Comparison between capture strategies: an n-shot burst captured with a longer focal length. The experiment shown in Figure 6.6 and Figure 6.7 is repeated with 2× focal length. Note that output images for AA contain more amount of blur. Also, a good exposure time for each approach is marked in (b), but this time, the exposure time for AA is reduced to half.

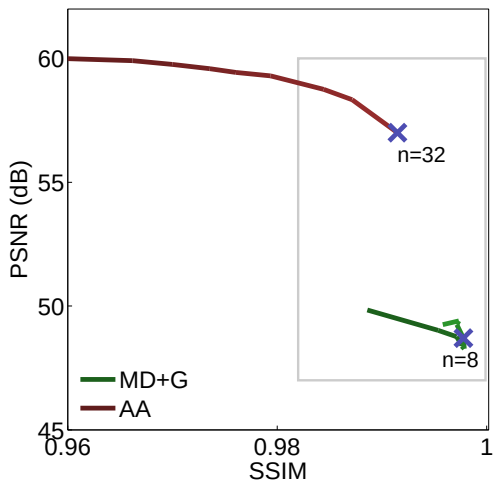
6.4.2 Simulation Results: Fixed Total Capture Time

Now let us consider the case when total capture time is fixed and user wants to determine how many images to capture during the time. This case is hard to analyze theoretically because the number of shots is also a variable. Again, we use our simulation pipeline to compare capture strategies. Figure 6.9 shows the results when the *Brasserie* scene is used and total capture time is fixed to 1s. We assume that the bandwidth of the camera system is high enough to capture 32 frames per second and the focal length is scaled by a factor of 2. Figure 6.9a shows that AA works well when n is large. Capturing more images results in shorter per-image exposure times, and thus, reduces the amount of handshake blur per image as well. In turn, the total amount of read noise accumulates with the number of captured images. The performance of MD+G is relatively stable. However, the output image becomes more noisy as more images are captured. The trade-off curves shown in Figure 6.9b and Figure 6.9c represent that MD+G is able to produce sharper images with slightly more noise. As long as n is large and captured images are well aligned, both approaches generate comparable results that exist in different points in the trade-off space. For this scene, taking $n = 8$ shots followed by MD+G or applying AA to a burst of $n = 32$ images can be considered as a good strategy, which is shown in Figure 6.9a and Figure 6.9b. Note that MD+G can generate good results with fewer images, compared to AA.

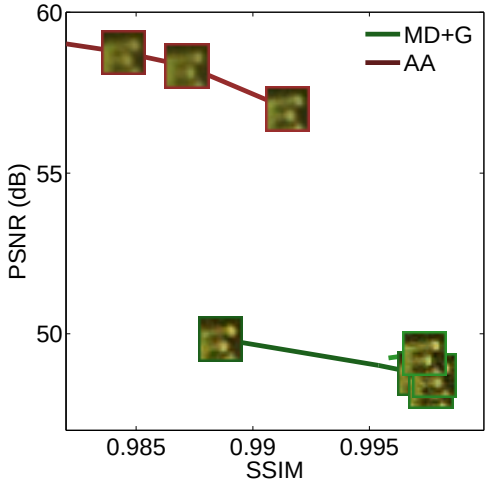
However, the choice may be different for darker scenes. For example, let us consider the *Tequila Bar* scene, which is considerably darker than the *Brasserie* as shown in Figure 6.5a. The simulation results for this scene are shown in Figure 6.10 and Figure 6.11. Note that output images in Figure 6.10a suffer from a higher noise level. When $n = 32$, our camera motion estimation failed due to the excessive amount of noise, and both approaches produced poor results. The output images in Figure 6.10b are obtained by repeating the simulation with ground truth camera motion. The difference between the two results shows how error in camera motion estimation affects the performance of two approaches. In these very low-light situations, it is not recommended to use the shortest exposure time to avoid blur. Instead, it is better to capture images with longer exposure times that allow accurate camera motion estimation, and then apply MD+G to remove handshake blur. Figure 6.11a and Figure



(a) Image insets

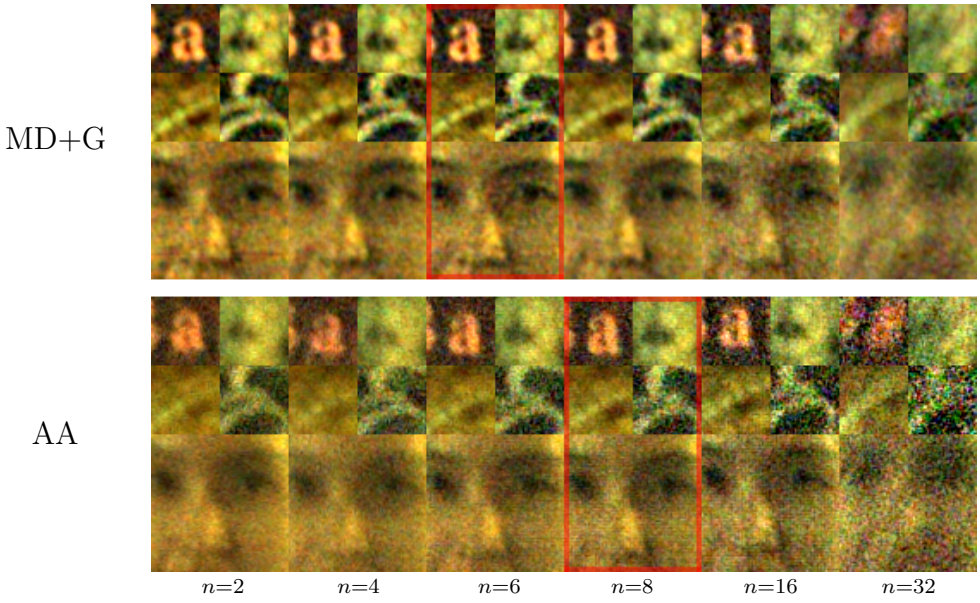


(b) Trade-off space

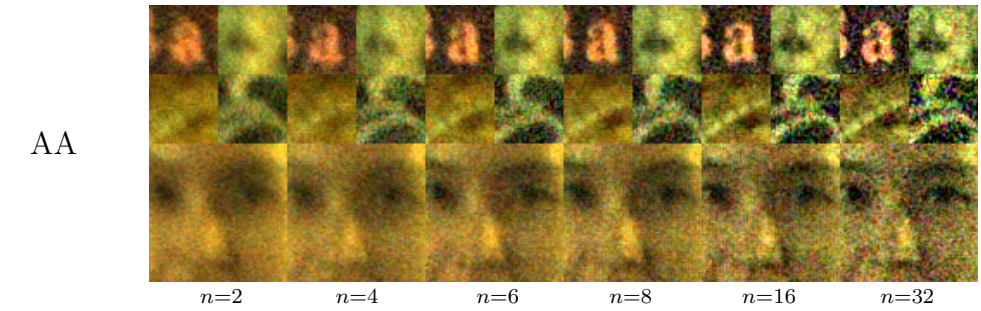


(c) Trade-off space w/ insets

Figure 6.9: Comparison between capture strategies: the *Brasserie* scene with fixed total capture time. The total capture time, which is set to 1s, is split between n shots with the same exposure time. AA gives slightly blurry images with less noise when n is large enough, while MD+G recovers more image details with more noise. As shown in (b) and (c), these two approaches provide results that lie in different regions of the trade-off space. A possible choice of n for each approach is marked in (b), and the corresponding output image is denoted in a red box in (a).

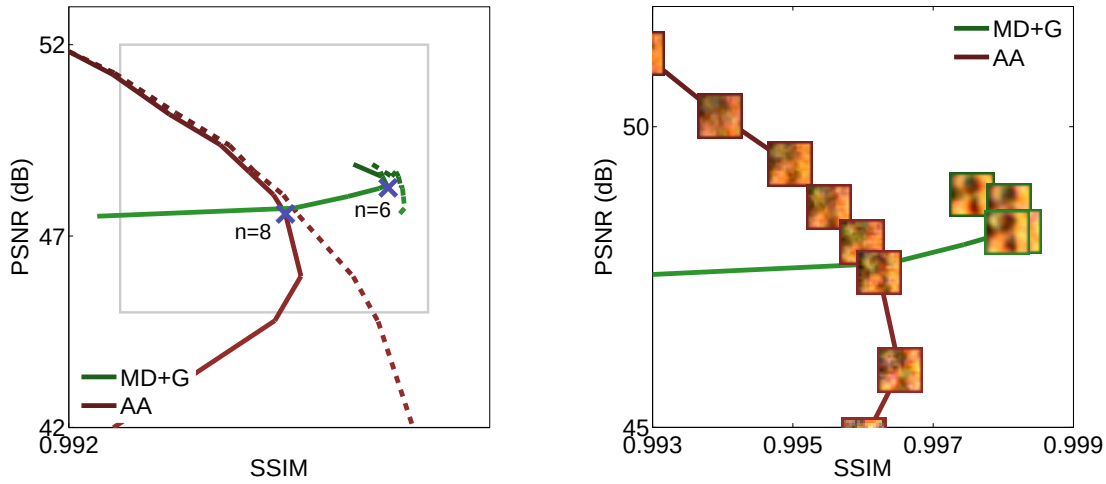


(a) Image insets, w/ our camera motion estimation



(b) Image insets, w/ ground truth camera motion

Figure 6.10: Comparison between capture strategies: the *Tequila Bar* scene with fixed total capture time (image insets). As shown in (a), both methods fail when n is large because our camera motion estimation cannot handle the excessive amount of noise. The output images shown in (b) are obtained by repeating the simulation with ground truth camera motion. The difference between these outputs demonstrates the importance of accurate camera motion estimation. The trade-off curves for this experiment are shown in Figure 6.11.



(a) Trade-off space (b) Trade-off space w/ insets

Figure 6.11: Comparison between capture strategies: the *Tequila Bar* scene with fixed total capture time (plots). In subfigure (a), two trade-off curves in solid lines represent the case when our camera motion estimation is used, while the curves in dashed lines correspond to the simulation results with ground truth camera motion. Although both methods fail for large n, MD+G can give better results than AA when smaller n is used and reach to the better part in the trade-off space shown in (a) and (b).

6.11b demonstrate that MD+G can reach the better part in the trade-off space.

6.5 Guidelines for Choosing Capture Strategies

Here we summarize some guidelines for choosing capture strategies that could be distilled from the various experiments preformed in the previous sections. Given a photographic situation, our goal is to determine how many images should be captured, what their exposure times should be, and which processing, AA or MD, should be applied.

To begin, let us list some important points that should be considered when AA or MD is used:

- **Align-and-average** There exists an upper bound in exposure time that gives tolerable amount of handshake blur.

- **Multi-image deblurring** Exposure time can be increased up to the point at which read noise is negligible and handshake blur becomes intolerable. Also, the number of images should be large enough to make deconvolution stable.

The amount of handshake blur is directly related to focal length of the optics. When a narrow field of view is used, the required exposure time to avoid blur may be too short to capture any meaningful information on the scene. Also, handshake blur depends on the type of camera and user. These issues should be considered to determine when AA can give reliable results. In contrast, MD is relatively more flexible with the choice of exposure time. However, there exists an upper bound in exposure time that benefits MD. This point comes at shorter exposure times if the scene is brighter or if more images are captured.

There are more rules that apply to both approaches. First, exposure time should be long enough to allow accurate camera motion estimation. Because accurate estimation is predicated on having low noise, there exists a lower bound on exposure time. This lower bound is affected by both scene brightness and scene content. Second, exposure level, which is the combination of exposure time and analog gain, should be carefully determined to avoid over-exposing bright regions. When a scene with wide dynamic range is to be captured, the range of proper exposure levels can be limited.

In addition, a few practical issues may limit the choice of capture parameters. For example, the effective range of analog gain is restricted by the presence of read noise added before analog amplification. Also, limitations in system bandwidth may bound the choice. For low-bandwidth camera systems, a blanking time may exist between exposures due to slow readout speed. This blanking time may significantly reduce the efficiency of capture because no photon is collected during the time. Further, the memory buffer in the camera system, which is used to store captured images before they are processed, has a limited size, restricting the maximum value of n. Finally, n is bounded by the total capture time during which the photographer can expect the scene to hold still.

In many situations, AA is preferred to MD because the former has lower computational cost. A good strategy is to find a parameter set that gives desired image quality with AA. However, it may not be possible for certain scenes, for example,

in low-light environments when read noise is dominant or analog gain has already been maximized. In these situations, MD can provide an alternative with better performance.

Chapter 7

Conclusion

This dissertation presents a new image deblurring system that combines gyro-based camera motion estimation and non-blind multi-image deconvolution. Our method tackles various challenges encountered in real-world scenarios to allow practical implementation in mobile systems. Then, a theoretical analysis on multi-image deconvolution provides insight into capture strategy selection. The key insights can be summarized:

Multi-image deconvolution is a robust method that is applicable to a wide range of low-light situations. When this method is used, exposure time can be increased up to a point when read noise is negligible and handshake blur becomes uncorrelated to each other.

In this final chapter, the limitations of our system are discussed, and we lay out a few possible directions to extend our system.

7.1 Limitations

The success of our method depends on how faithful our models for image formation and blur representation can explain the real measurements from image sensor and gyroscope. In practice, the assumptions we made for the models are often violated

and these violations are the main cause of degradation. Here, we list a few cases.

First, the translational component of camera motion may not be negligible. When objects are placed close to the camera, or when the user moves the camera around the scene, gyroscope data cannot provide accurate estimation anymore. During the gyroscope drift correction stage, the search of proper gyroscope bias is done in a constrained space that is defined by our rotational model. When translation matters, we might relax our model and search for image translations as a refinement step. Although full motion cannot be recovered, this additional step will find the best camera motion that explains the captured images.

Second, obtaining accurate timestamps for each exposure time window is not trivial. Although the length of exposure time can be considered deterministic in many camera systems, the blanking time between exposures is not easy to find because there are unexpected time delays caused by the processing pipeline and occasional frame drops. Frame drops are one of the main causes of poor results in our implementation. If we allow exposure times to vary during the capture, then reconciling timestamps becomes more complicated and is specific to the camera system.

Finally, when a very short exposure time is used with a longer frame time, our camera motion estimation may suffer from large gaps between frames. When long frame times are used in bandwidth-limited systems, neighboring image frames are less correlated to each other, and our estimation is more likely to stay in local minima. When images are sharp with tolerable amount of noise, image-based alignment methods can be a superior alternative.

7.2 Further Applications

Here are a few more possible directions to extend our gyro-based multi-image deblurring method. Our method can be combined with other applications to make it more robust to tackle real-world situations.

7.2.1 Removing Blur in Moving Objects

Our method only removes blur in static regions. Although the errors in moving objects do not cause severe artifacts in the multi-image approach, these objects remain blurry after deblurring.

Our approach can be combined with other motion deblurring methods [5, 13] to remove local motion blur that exists in images captured from hand-held devices. The motion of moving objects can be separated from global blur caused by camera shake, which is measured by a gyroscope. As shown in Section 4.4, our camera motion estimation is robust to the existence of moving objects in the scene because static regions, which appear to be sharper than moving objects, guide our gyroscope bias estimation. Then, moving objects can be detected by Equation 4.5 and treated separately with blind motion deblurring methods. This approach can be extended to video data as well.

7.2.2 Gyro-Based High Dynamic Range Photography for Hand-held Devices

In low-light situations, the use of multi-image deblurring methods is often restricted by their limited capability in handling wide dynamic range scenes. Longer exposure images can improve output images, but may also blow out interesting image features that exist in bright regions.

Our gyro-based multi-image deblurring method can be easily extended to tackle this problem. Instead of capturing a burst with the same exposure parameters, we can deliberately vary exposure times among captures. A good capture strategy would be to take an image with short exposure time to preserve highlights and capture multiple images with long exposures to obtain dark regions, but with some blur. Based on the camera motion estimated from gyroscope data, these images are jointly deblurred to recover a sharp HDR image. Our multi-image deblurring, which is described in Equation 5.1, is not limited to handling images captured with the same exposures. With a small modification to handle over-exposed regions in long exposures, our method can be applied to an HDR burst.

This idea is similar to Lu et al. [41]. Our method improves their work by using gyroscope measurements. Also, it would be interesting to design a new HDR metering method for hand-held captures. Our analysis on the performance of capture strategies can help selecting capture parameters that are expected to give good results.

7.3 Closing Remarks

The goal of computational photography is to extend the capabilities of digital photography. In particular, this dissertation focuses on obtaining good images with hand-held devices in low-light situations where a traditional camera fails. In such situations, an attached gyroscope can be used to measure camera shake and then computationally recover high-quality images by combining blurry images captured from a burst capture. We hope that this gyro-based multi-image approach enables future computational photography applications that run on mobile devices.

In addition, this dissertation analyzes how good capture strategies can be determined for hand-held image captures. We hope that key insights provided in this dissertation inspire camera software designers and researchers in the field of computational photography to tackle various challenges in hand-held digital photography.

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